

Bayesian Gene Ontology Benchmark Dose Estimation Supplement

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1 Bayesian Model Implementation Details

The full hierarchical model is indexed by gene i , latent factor dimension h with $1 \leq h \leq m$, and dose observation j . Doses are referred to as observations because they have replicates. Recall that λ_i represents the i th row of the matrix Λ , corresponding to the i th gene loadings.

$$\begin{aligned} Y_{ij} &= X_j \beta_i + \lambda_i \eta_j + \epsilon_{ij} \\ \beta_i | c_i &\sim N(0, I c_i^{-2}), \quad \epsilon_i | \tau_i \sim N(0, I \tau_i^{-2}) \\ \eta_{hj} &\sim N(0, 1) \quad \lambda_{ih} | \gamma_h, \phi_{ih} \sim N(0, \phi_{ih}^{-1} \gamma_h^{-1}) \end{aligned}$$

This specification is agnostic to any gene set; each parameter is either gene specific or latent factor specific. Hyperpriors are specified with fixed hyperparameters:

$$\begin{aligned} c_i &\sim \Gamma(1, 1), \quad \tau_i \sim \Gamma(2, 2) \quad \phi_{jh} \sim \Gamma(v/2, v/2) \\ \gamma_h &= \prod_{\ell=1}^h \delta_\ell, \quad \delta_1 \sim \Gamma(a_1, 1), \quad \delta_\ell \sim \Gamma(a_2, 1) \end{aligned}$$

The covariate matrix X is generated from a B-spline basis with four basis functions (ie four knots) evaluated at the observed doses on the log scale. For creating figures with smooth curves, the basis functions are re-evaluated at the same knots but a finer scale over the dose range. Since the basis functions are not changed, the fitted coefficients can be used to plot the fixed effects or estimate where the test statistic crosses the BMR-based threshold via interpolation. Given an initial m possible factors, the sparsity inducing prior for λ shrinks unneeded factors to 0. Here we choose the latent factor dimension m to be 15 and typically only 2-5 dimensions remain after shrinking.

2 MCMC Diagnostics

2.1 MCMC Trace Plots

We give evidence that our MCMC fitting procedure rapidly converges to the posterior. Given the model’s complexity, we emphasize that the model parameters, especially the latent factor terms, can appear to not converge but still result in stationary distributions for the computed test statistics. This is often due to rotational symmetries and individual model parameters being non-identifiable; consequently, we focus our diagnostics on identifiable quantities that have actual meaning (e.g., correlation, and the BS-BMD).

Figures 1 and 2 show trace plots of the benchmark doses for two hallmark sets. The BMD reaches stationarity within 1000 iterations or the BMD values concentrate on the maximum dose with occasional samples appearing in the dose range. The supplement shows the Gelman-Rubin diagnostic for stationarity for all the parameters and suggests that the posterior samples are stationary. The results are based on 7000 posterior samples. Assuming a burn-in of 2000 as presented, the effective sample size for the BMD estimates for each gene set vary from 24% to 65%, meaning the 5000 MCMC samples corresponds to between roughly 1200 and 3200 independent samples. When there is no signal, the posterior sample consists of just the maximum dose and the effective sample size is not well defined.

2.2 Stationarity

The full data set of Gwinn et al. [2020] contains 18 chemicals measured from two tissues and 50 hallmark gene sets for a total of 2000 BMD estimates. For each of those estimates, our MCMC procedure was run for 7000 iterations and the first 2000 were discarded as burn in. The resulting 5000 iterations were then tested for the effective sample size using the partial autocorrelation, which ranged from 0.2356 to 0.6461. This implies the 5000 iterations had an effective sample size of 1178 to 3230.

Stationarity of the MCMC chain is checked with the Gelman-Rubin diagnostic, which we compute in R using the coda package with parameter chains from four separate runs. The diagnostic values for the latent factors such as Λ and η were not consistent across chains, which is expected because the individual

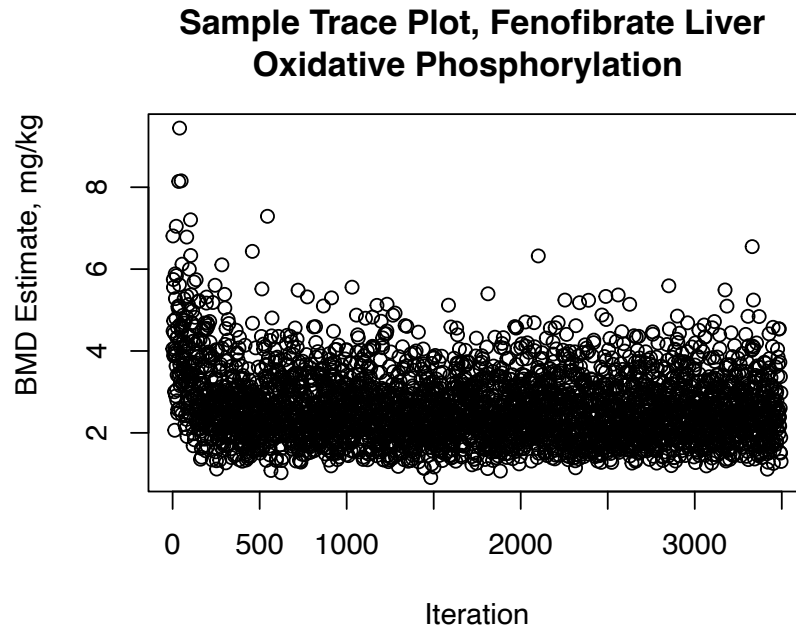


Figure 1: Trace plot showing sampled values for the BMD for the oxidative phosphorylation set under Fenofibrate in the liver. Only half of the 7000 sampled points are shown for clarity. Stationarity is achieved within 1000 iterations

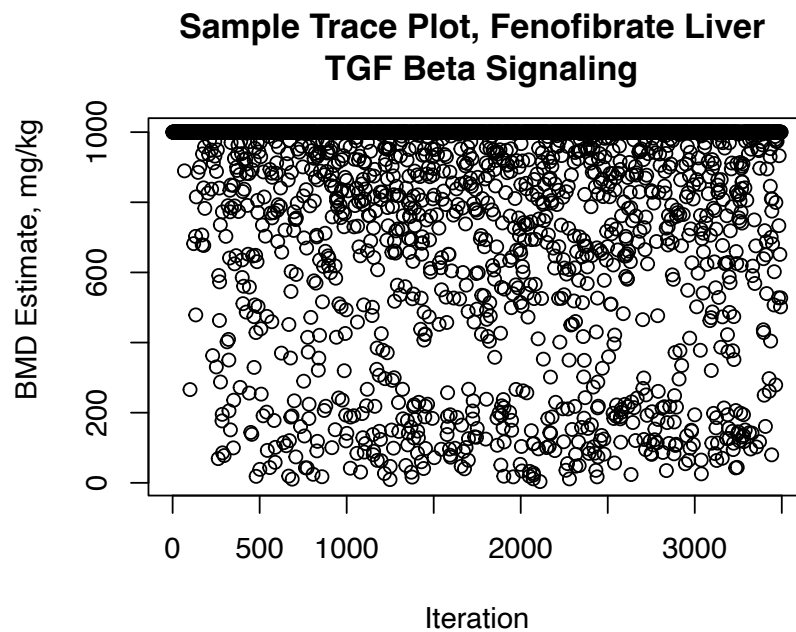


Figure 2: Trace plot showing sampled values for the BMD for the TGF Beta signaling gene set under Fenofibrate in the liver. Only half of the 7000 sampled points are shown for clarity. The concentrated mass on the maximum dose is typical for gene sets with weaker signals.

parameters are not identifiable. However, the aggregated parameters in the form of the covariance or BMD estimate were confirmed to have satisfactory agreement and stationarity across chains. Since there are millions of parameters in the covariance, we only provide summary statistics for a subset of the first 50 genes (for 2500 covariance parameters) for fenofibrate liver and kidney and assume that the rest of the parameters and chemicals have similar behavior.

Test Parameters	Size	Median	Max
Covariance (Liver, genes 1:50)	2500	1.00	1.14
BMD (Liver)	50	1.00	1.04
Covariance (Kidney, genes 1:50)	2500	1.05	1.51
BMD (Kidney)	50	1.00	1.09

Table 1: Diagnostics for Fenofibrate Liver. There are 50 hallmark sets but over 3 million unique covariance parameters, so we only test a subset of 2500 covariance parameters.

3 Probabilistic Formulation

Here we review the derivation of Crump [1995] from a probabilistic perspective and observe that BMR is defined as both a “fold-change” scalar and as a change in the probability of an adverse response in that work. Let the cumulative distribution function for the response R given a centered response be denoted by F_R and the inverse CDF by F_R^{-1} . By definition, we have

$$P(R > F_R^{-1}(0.99)) = 0.01$$

The method of Crump [1995] is then to find a threshold H where the adverse response has increased by a specified Benchmark Response (BMR) such as 10%,

$$P(R + H > F_R^{-1}(0.99)) = 0.01 + \text{BMR} = 0.11$$

Moving the constant H to the right side, we have

$$P(R > F_R^{-1}(0.99) - H) = 0.11$$

An immediate solution is to let $H = F_R^{-1}(0.99) - F_R^{-1}(0.99 - \text{BMR})$, which is approximately the 1.1 value mentioned earlier when $F = \Phi$, the Gaussian CDF. This expression is easily generalized to higher dimensional cases by substituting for the new random variable of interest. For example, aggregating with squared, normalized terms such as $\hat{R} = \sum_i R_i^2 / \sigma_i^2$ yields the same basic calculation,

$$P(\hat{R} + H > F_{\hat{R}}^{-1}(0.99)) = 0.01 + \text{BMR}. \quad (1)$$

As before, we can use the trivial result $H = F_{\hat{R}}^{-1}(0.99) - F_{\hat{R}}^{-1}(0.99 - \text{BMR})$. Using a 10% change in adverse response for the BMR will not recover identical statistics for all transformations, so care is needed to match the effect of the univariate case. Additionally, if γ is the extreme single-tail probability of interest, then this generalizes to $H = F_{\hat{R}}^{-1}(1 - 2\gamma) - F_{\hat{R}}^{-1}(1 - 2\gamma - 2\text{BMR})$, for our two-tailed approach. As described in the main text, the quadratic response formula requires doubling the probability mass to account for the negative tail being folded into the positive tail. This procedure assumes that the error distribution does not change with the center, which is not generally true. However, the observed doses may not be sufficiently dense to provide error distributions near the true BMD, so the assumption of equal variance (homoskedasticity) is a reasonable simplification and alternative to attempting to interpolate the error distribution between doses.

4 BMD analysis using BMDExpress

The transcriptomic BMD analysis using BMDExpress 2.30.0512 was performed according to the US EPA’s Standard Methods for Development of ETAPs (<https://doi.org/10.23645/epacomptox.25365496>). Data were

loaded into BMDEpress and gene annotations dated 01/07/22 were applied. Loaded data underwent a prefilter to remove probes that did not exhibit a significant dose-related monotonic change using a Williams trend test (0.05, 10000 permutations) and a minimal change of 1.5-fold (treated at any dose level vs control). Probes which passed the prefilter were subject to benchmark dose modeling using the EPA BMDS Models. Models used included Hill, Power, Linear (Polynomial 1), Polynomial 2, and Exponential M2, M3, M4, M5. The BMR level was set to 1.349 standard deviations (relative to the fit at control). To identify the best-fit model for each fitted probe, the Akaike information criterion (AIC) values for each fitted model were compared and the model with the lowest AIC was selected. The best model for each probe was used to calculate the BMD, BMDL, and BMD upper confidence limit (BMDU).

The specific parameter settings selected from the BMDEpress software when performing probe-level BMD analysis were as follows: maximum iterations: 250, confidence level: 0.95, BMR factor: 1.349, restrict power: no restriction, and constant variance: selected. The specific model selection setting in the BMDEpress software when performing probe set-level BMD analysis was as follows: best poly model test: lowest AIC, flag Hill model with “k” parameters: $< 1/3$ the lowest nonzero dose tested, and best model selection with flagged Hill model: select next best model.

To group dose-responsive genes into Hallmark gene sets, the defined category analysis in BMDEpress was employed. Before probes were transformed into genes and passed into the Hallmark gene sets, they were filtered to remove probes that did not meet criteria for acceptable qualities. Specifically, probes were removed if (1) they had a BMD greater than the highest dose tested, (2) they mapped to more than one gene, (3) they had a global goodness-of-fit p-value ≤ 0.1 , or (4) with a BMD/BMDL ratio > 20 . Probes that passed these criteria were mapped to genes and passed into the Hallmark gene sets. Hallmark gene sets were considered active if they contained at least 3 genes that showed dose-responsive

5 Null Hypothesis Ambiguity and Significance

We do not conduct formal hypothesis testing when determining if a gene set shows risk. A null distribution for a BMD is more complicated to define than a null distribution describing the different expression between two sets. First, the BMD as presented is intended to measure an absolute risk rather than a relative risk. While one could develop a test to see if two sets have a significantly different BMD, the regulatory question of deciding if the individual BMDs present a risk still remains. The fact that BMDs from two sets are not significantly different from each other doesn’t help unless we know that one BMD is not significant to begin with.

Additionally, there is a conflict between a biological null and a mathematical null. The biological null suggests that for any substance, a sufficiently high dose will be hazardous, but not necessarily for the reasons represented by the gene pathways. The mathematical null suggests that, focusing just on the pathway of interest, even an infinite dose will not cause a problem because the chemical just doesn’t affect the genes. The latter null suggests a distribution with mass at infinity while the former null suggests a vague prior with probability mass only at finite but very high values. It is not clear how to balance these nulls, nor is it clear how to specify either distribution.

Furthermore, the experiment itself is censored. If a BMD is found to be at or above the highest dose, it is perhaps most correct to conclude that the experiment is insufficient to determine significance, rather than accepting a null hypothesis that the gene set shows no response. From a regulatory point of view though, it is clearly evidence that the chemical does not pose a risk at the tested doses, so setting the BMD to the maximum dose is a useful signal.

One could attempt to infer the null BMD from the null on the gene expression. Our modeling approach presumes that the change in response given a change in dose should be zero:

$$E(\mu(d) - \mu(0)) = 0$$

The responses are assumed to have Gaussian error but may have different means and be correlated. This is similar to the class 3 null described by Barry et al. [2008] which is recommended for testing to avoid false positives or excessively conservative p-values for functional groups of genes. This null allows a statistician to test if a random gene set shows expression that is significantly different from an arbitrary gene set (which may have correlated signals by chance and shouldn’t be assumed to have zero signal), and could in principle

be used to determine if the BMD of a gene set is significantly different from the BMD of an arbitrary set of equal size. However, this test doesn't elucidate if the original gene set expression represents a biological risk.

In light of these issues, we emphasize that the risk assessment of a BMD is fundamentally different from a hypothesis test. Although both risk evaluation and testing involve tail probabilities, we are not testing if gene sets are differentially activated compared to a random set from the genome. Instead, we check if a gene set has an aggregated response that is above its own background noise enough to satisfy a predetermined risk level. Crucially, we rely on the correctness and relevance of the gene set. The fact that the gene set may exhibit a risk by chance inclusion of responsive genes is not considered a false positive because the risk is not a comparison to the genome.

Although we make distributional assumptions about the response and our risk statistic is an approximation, a future direction is to apply non-parametric methods similar to what is described in Barry et al. [2008] to avoid the distributional assumptions. Such an approach allows one to estimate a response distribution to estimate the BMD. We leave this problem to future research.

6 A Brief Simulation Study

A gene expression simulation study is difficult to perform because a "realistic" simulation is not well defined. As an approximation, we can take a real set of gene expressions and remove any dose responses without any further centering or scaling. Empirically, this approach recovers a sufficiently rich set of data in terms of preserving gene correlations. (Type 3, Barry et al. [2008]).

6.1 Single Active Gene Study

We perform a simple simulation study to test if the size of a gene set affects the BMD estimate. Given the baseline data from the procedure above, we activate a single gene in each Hallmark set by adding a predetermined sigmoidal signal with fixed magnitude and inflection point: a Hill function with sill or maximum effect value of 2, slope 1, and $EC_{50} = 100$. While we could generate a sequence of arbitrary gene sets, seeing the effect on the hallmark sets can provide more insight for the application results.

The results of this simulation study in Figures 3 and 4 unsurprisingly, suggest that the BMDs for small gene sets are more sensitive to a single gene having a response. The effect seems to diminish for the BMDL once the gene set is above roughly 50, which could be explained by the aggregate noise of the larger gene sets washing out the lone gene signal. Table 2 illustrates that the order is not completely consistent with gene set size. These results highlight the need for proper communication between the modeler and the biologist, and it is not necessarily a problem, from a risk point of view. Intuitively, a single gene's activation will more easily perturb the background distribution from a small set compared to a pathway with hundreds of genes. Determining if the change is important biologically is a qualitative decision in a risk assessment and not a quantitative one. Our approach provides a method for evaluating *a priori* sets; it does not provide a method for creating the sets themselves.

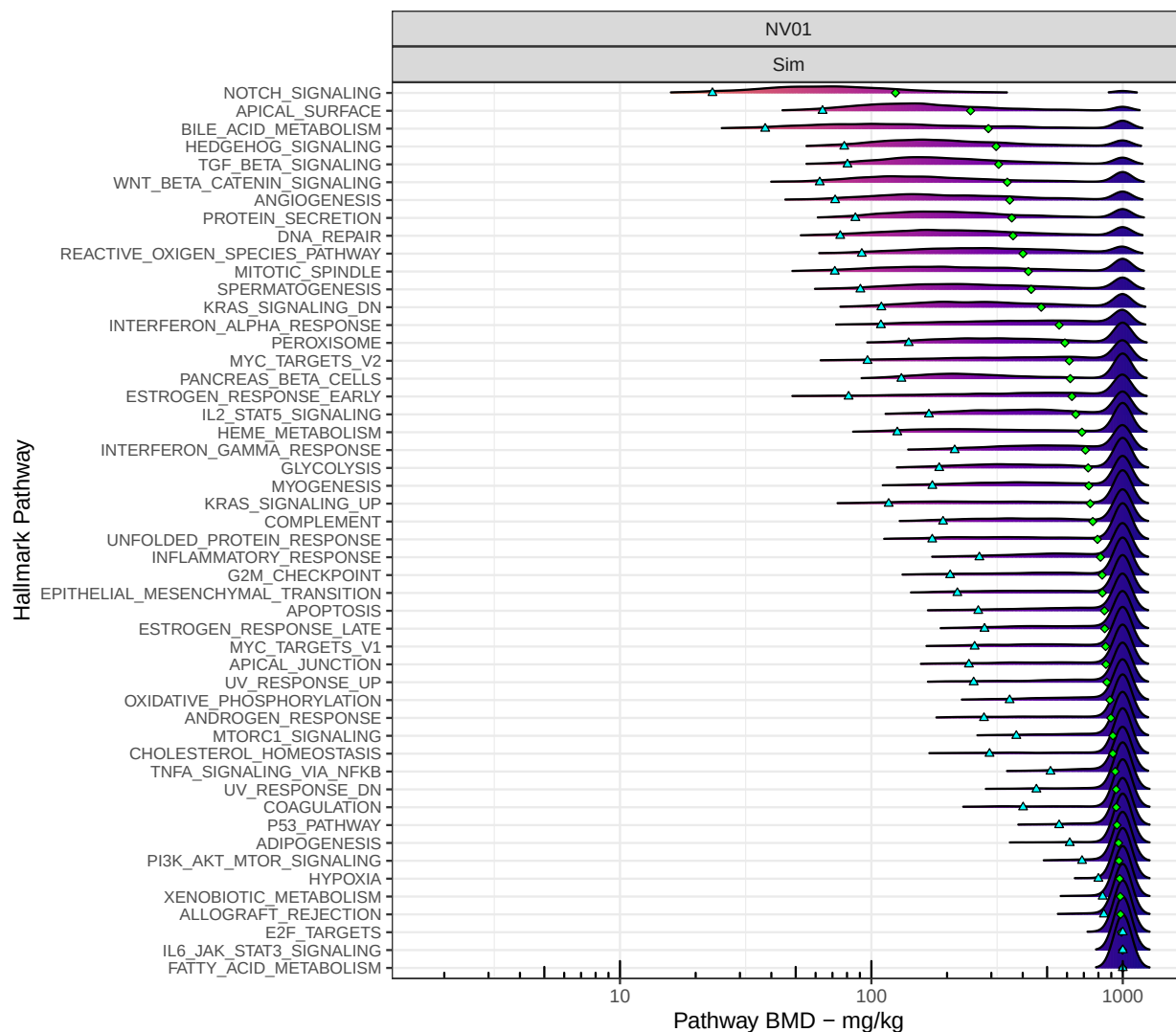


Figure 3: A simulation study where a single gene is activated in each Hallmark set. The base data is taken from 5-day study and dose responses are initially removed while preserving correlation. The added signal is a Hill function sigmoid with sill or maximum effect of 2 and EC50 of 100 and is only added to one gene in each Hallmark set. The resulting BMD values roughly reflect the size of the gene set, but even large gene sets show risk through a lowered BMDL.

Table 2: The simulation study suggests that small gene sets are more likely to have a small BMD and BMDL when a single gene is activated in a set. The ordering by Total genes is loosely, but not exactly, consistent with the ordering by BMDL.

	Hallmark	Total Genes	Percent Active	BMD	BMDL
1	Notch Signaling	9	11	124.7	23.32
2	Pancreas Beta Cells	9	11	618.2	131.64
3	Apical Surface	13	8	248.0	63.93
4	Hedgehog Signaling	14	7	313.5	78.00
5	Wnt Beta Catenin Signaling	17	6	347.3	62.29
6	Tgf Beta Signaling	17	6	320.8	80.33
7	Angiogenesis	20	5	355.0	71.74
8	Protein Secretion	24	4	361.7	86.37
9	Myc Targets V2	24	4	612.6	96.52
10	Reactive Oxygen Species Pathway	23	4	400.7	91.59
11	Cholesterol Homeostasis	38	3	913.3	295.07
12	Il6 Jak Stat3 Signaling	31	3	998.5	1000.00
13	Dna Repair	31	3	366.0	75.17
14	Androgen Response	39	3	896.1	280.47
15	Interferon Alpha Response	38	3	558.2	109.09
16	Heme Metabolism	38	3	687.9	126.79
17	Bile Acid Metabolism	37	3	292.1	37.81
18	Spermatogenesis	34	3	432.1	90.41
19	Kras Signaling Dn	33	3	473.7	109.57
20	Mitotic Spindle	43	2	421.2	71.52
21	Adipogenesis	63	2	962.3	615.08
22	Estrogen Response Early	60	2	627.5	81.17
23	Apical Junction	61	2	857.2	244.27
24	Unfolded Protein Response	46	2	793.2	174.56
25	Pi3k Akt Mtor Signaling	56	2	965.9	688.38
26	Myc Targets V1	55	2	854.8	257.41
27	Inflammatory Response	65	2	815.3	268.84
28	Oxidative Phosphorylation	40	2	889.0	354.71
29	Glycolysis	66	2	728.3	186.13
30	Uv Response Up	57	2	864.4	255.14
31	Uv Response Dn	51	2	940.8	453.30
32	Coagulation	66	2	941.8	401.26
33	Il2 Stat5 Signaling	64	2	650.1	169.36
34	Peroxisome	42	2	588.5	140.71
35	Kras Signaling Up	60	2	741.9	117.15
36	Tnfa Signaling Via Nfkb	88	1	933.1	515.69
37	Hypoxia	86	1	972.1	799.66
38	G2m Checkpoint	84	1	828.0	205.79
39	Apoptosis	91	1	845.4	266.16
40	Estrogen Response Late	70	1	847.5	281.81
41	Myogenesis	84	1	732.5	174.75
42	Interferon Gamma Response	78	1	710.4	214.44
43	Complement	85	1	759.8	192.76
44	Mtorc1 Signaling	98	1	913.3	377.17
45	E2f Targets	94	1	988.4	1000.00
46	Epithelial Mesenchymal Transition	85	1	829.9	219.76
47	Xenobiotic Metabolism	94	1	977.1	831.91
48	Fatty Acid Metabolism	72	1	999.1	1000.00
49	P53 Pathway	85	1	948.2	558.30
50	Allograft Rejection	78	1	979.5	841.13

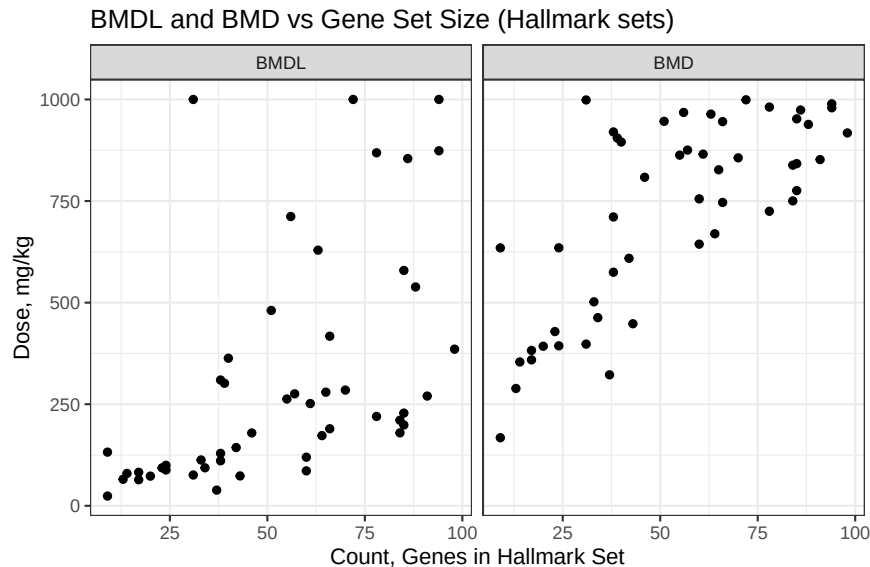


Figure 4: The results of the simulation study suggest that there is a correlation between the gene set size and the BMD when a single gene is activated with a fixed signal. However, the correlation for the BMDL becomes quite weak as the gene set size increases, suggesting that the fixed signal is getting washed out by the aggregate noise in the larger gene sets.

7 Detailed Analyses for Showcased Chemicals

As noted above the ETAP methodology is focused solely on identifying a gene set PoD that aligns approximates a POD from guideline toxicological assessment. Often the gene set that is identified using this approach has limited biological plausibility based on what would be expected from prior knowledge of the toxicological/pharmacological properties of the chemical and hence limits the utility of the data to deriving a potency. To provide a comparison of methods we evaluate the dose-related gene set changes identified by the EPA ETAP approach and BS-BMD using the Hallmark gene sets. Admittedly this evaluation is somewhat subjective, hence we have selected 3 chemicals with well define mechanistic properties in the liver, fenofibrate, furan and methyleugenol with the goal of being as objective as possible. Due to space limitations the results for furan and methyleugenol are provided here in the supplement.

7.1 Fenofibrate: A Strong Signal Overcoming Correction

In Figure 5 we show a single posterior estimate for a single hallmark set for a well defined BS-BMD: the fatty acid metabolism hallmark gene set from the Fenofibrate analysis. With such a strong signal, correcting for variability from the model flexibility only shifts the BS-BMD across a small region.

The subsequent figure, Figure 6, illustrates how Hallmark set ordering is improved with the BS-BMD method.

7.2 Furan

Figure 7 shows samples from the posterior BS-BMD distribution for exposure to Furan. Furan is a volatile, colorless liquid often used in chemical manufacturing. If ingested or inhaled, furan is rapidly absorbed. In the liver it undergoes metabolic activation to an epoxide form primarily via the enzyme CYP2E1 [Gates et al., 2012], which then reacts with and damages cellular macromolecules such as protein and DNA [Robles, 2014]. This leads to activation of cellular responses to counter the effects of the reactive chemical (e.g., up-regulation of glutathione pathways and unfolded protein responses). Once counter measures are exhausted cell death begins to occur [Carfagna et al., 1993] and the activation of damage response signaling in the form

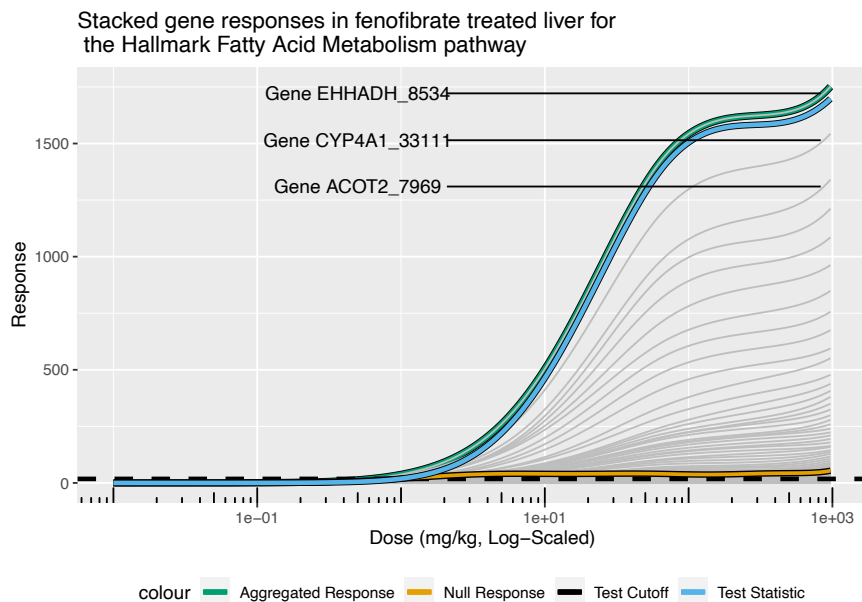


Figure 5: Strong signal in the liver with Fenofibrate. The Fatty Acid Metabolism hallmark set has 72 genes, resulting in a population noise distribution with larger mean and spread. The difference in response between the 98th and 78th quantiles is roughly 125, represented by the black dotted line. The aggregated dose response in solid black easily exceeds the limit and there are many genes that contribute to the overall response. The top three genes are shown and do not significantly vary across iterations.

of dedifferentiation [Kotulkar et al., 2023], inflammation [Robinson et al., 2016] and regenerative processes [Gilgenkrantz and de l’Hortet, 2018] are activated. In agreement with the typically steep dose response seen with furan [Moser et al., 2009] a sizable number of the gene sets in liver are activate in narrow window (BMDL 2-8 mg/kg/day) with the gene sets related to cell cycle and regenerative/proliferation responses showing the highest potency (E2F Targets, G2M checkpoint, Mitotic Spindle, MYC Targets 1 and 2). This is likely due to onset of proliferative lesions after 5 days of exposure to furan. Proliferation associated gene set responses were closely followed in potency by gene sets related to cell death (apoptosis) and dedifferentiation (fatty acid and xenobiotic metabolism, both down-regulated upon review of the data). Other notable responses in the very narrow gene set response range are reactive oxygen species pathway (related to glutathione metabolism), unfolded protein response, UV response up/DNA repair/P53 response (a related to response to damage to DNA) and inflammatory response. Consistent with the known weaker toxicological effects of furan in the kidney [Von Tungeln et al., 2017], no gene sets were found to be active in this organ. For both tissues, the set representing the entire genome of roughly 2500 genes, FULL GENOME, is in agreement with the hallmark sets, illustrating how our method does not wash out signals as the gene set size increases. Figure 8 shows improvement in Hallmark set ordering when using BS-BMD over the ETAP method.

7.3 Methyleugenol

Methyleugenol, a natural compound found in various essential oils and commonly used spices such as nutmeg [Tan and Nishida, 2012], belongs to a subclass of phenylpropenes called alkenylbenzenes. A number of these compounds are known to be genotoxic and hepatocarcinogenic in rats [Eisenreich et al., 2021]. Methyleugenol undergoes bioactivation primarily in the liver and is metabolized by cytochrome P450 enzymes to form reactive metabolites, such as 1’-hydroxymethyleugenol [Gardner et al., 1997]. These metabolites are capable of causing oxidative stress, depletion of glutathione [Yao et al., 2016], formation of DNA adducts [Phillips et al., 1984], and elicitation of mutagenesis [Jin et al., 2013]. This leads to organ damage, which secondarily results in proliferative responses and inflammation. If these effects persist, they can lead to liver cancer. Based on what is known about the toxicological effects of methyleugenol in the liver, it is expected that

Fenofibrate

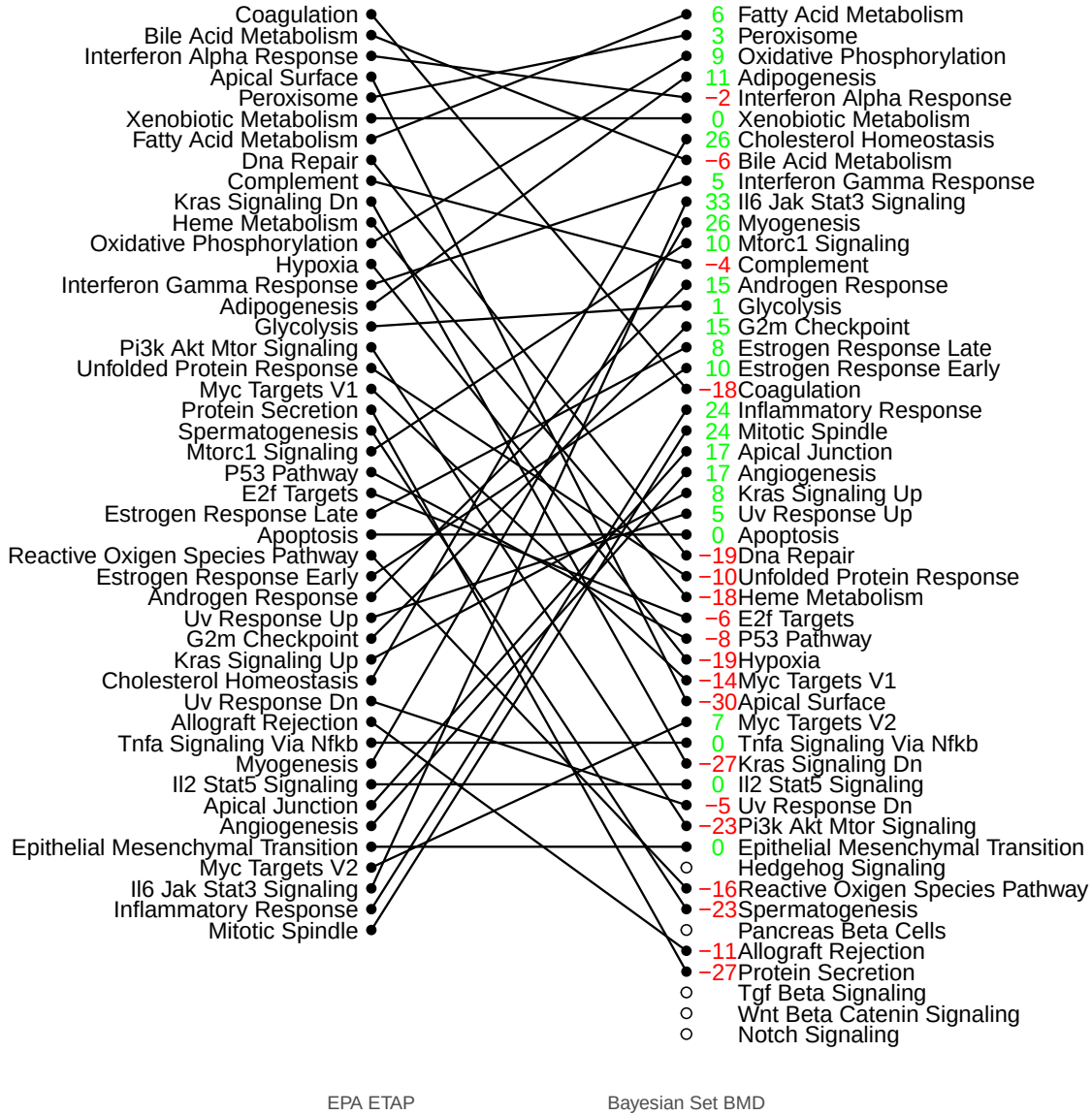


Figure 6: The hallmark sets are ordered by BMDL for fenofibrate exposure in the liver. Expert knowledge suggests that hallmark sets like fatty acid metabolism should be most impacted by fenofibrate in the liver, but the EPA ETAP method prioritizes less relevant sets. Numbers indicate difference in rank of hallmark set BMDLs between the two methods.

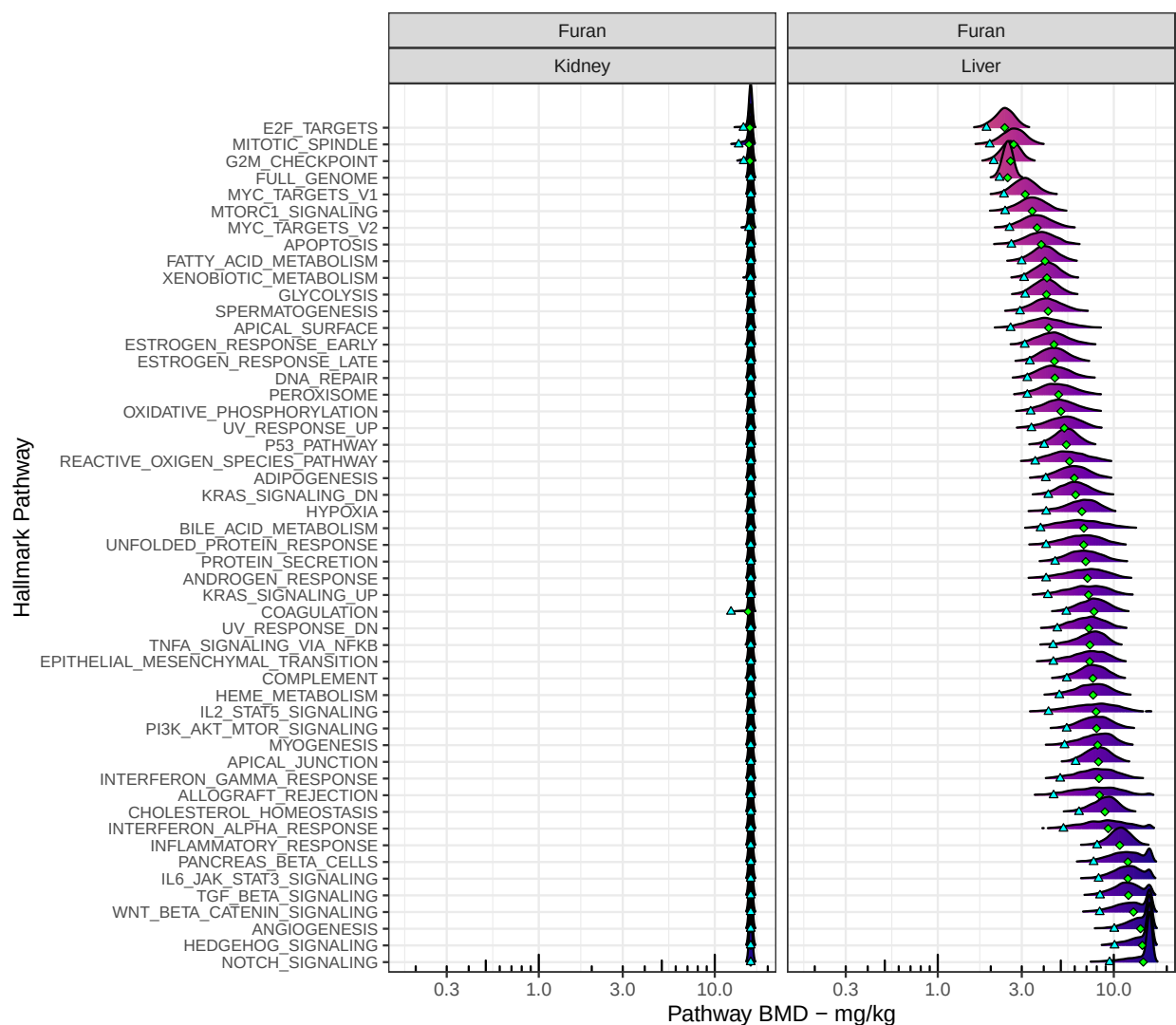


Figure 7: Furan BS-BMD simulations by each Hallmark pathway for kidney (left) and liver (right). The distribution of 5,000 posterior samples for the BS-BMD (X-Axis) are shown for each Hallmark gene set (Y-axis) as well as the median (diamonds), and 5th percentile (triangles). The maximum dose is returned as the BMD when the response is not significant, so the kidney shows no active pathways. The density is shown when the probability mass is at least 0.01.

Furan

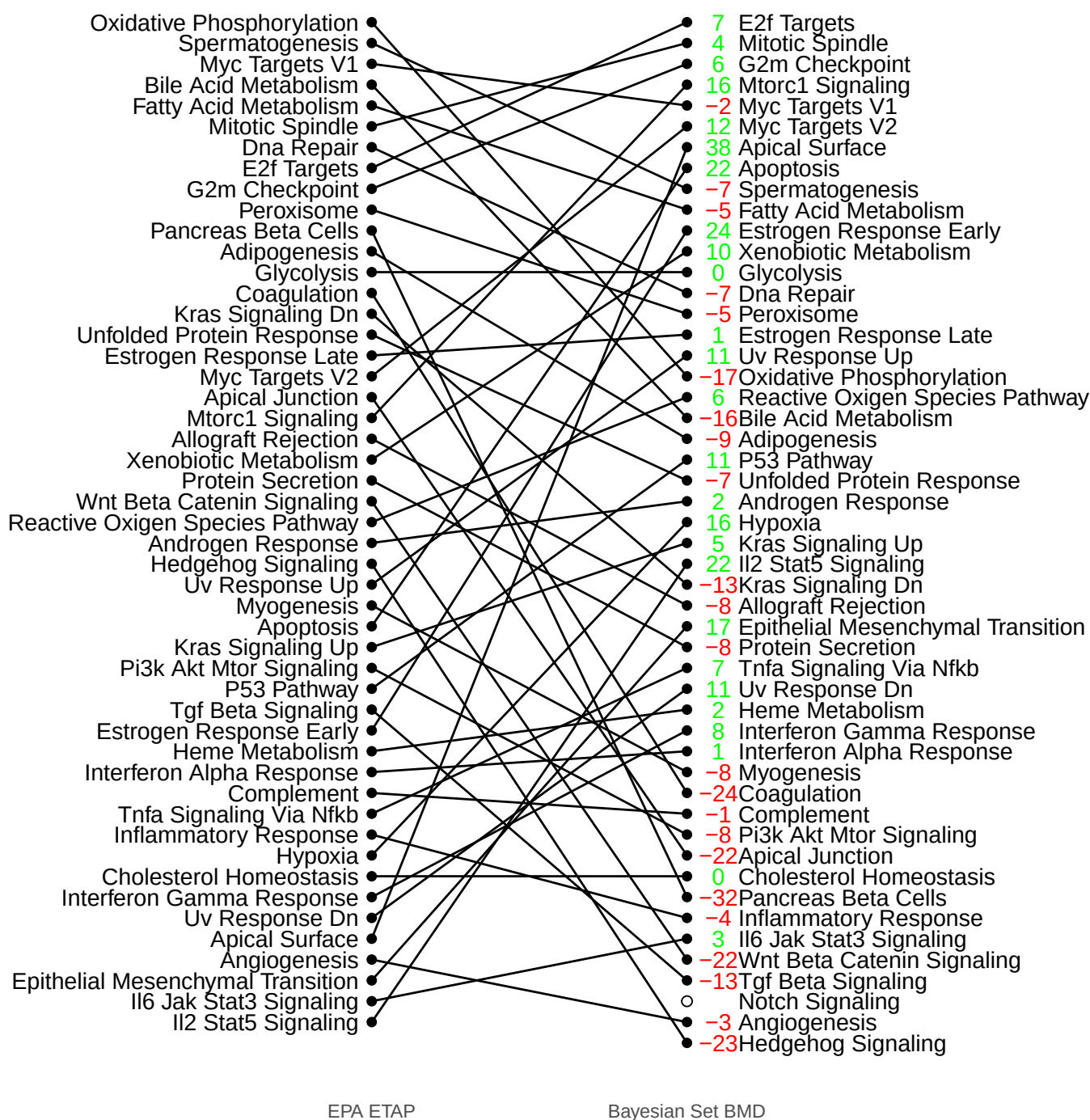


Figure 8: The hallmark sets are ordered by BMDL for furan exposure in the liver. Expert knowledge suggests that hallmark sets like related to cell cycle and regenerative/proliferation responses should be most affected, like E2F, MYC, and mitotic spindle, rather than a pathway like bile acid metabolism found by the EPA method. Numbers indicate difference in rank of hallmark set BMDLs between the two methods.

the most potently responding hallmark gene sets would be related to xenobiotic metabolism (response to the initial chemical insult), reactive oxygen species pathway (response to oxidative stress and glutathione depletion), P53 pathway (activated due to DNA damage), and E2F/Myc targets (cell proliferation). A review of the most potently responding hallmark gene sets according to the BS-BMD analysis includes xenobiotic metabolism, P53 pathway, and Myc/E2F targets. Interestingly, there also appear to be potent effects on two related gene sets, fatty acid metabolism and peroxisome, which may indicate a stress-related shift in essential liver functions, potentially a byproduct of damage caused by methyleugenol [Tarlow et al., 2014]. Similar to the results of furan, the FULL GENOME gene set agrees with the hallmark sets, illustrating how our method can correctly scale with gene set size without losing the signal.

Methyleugenol

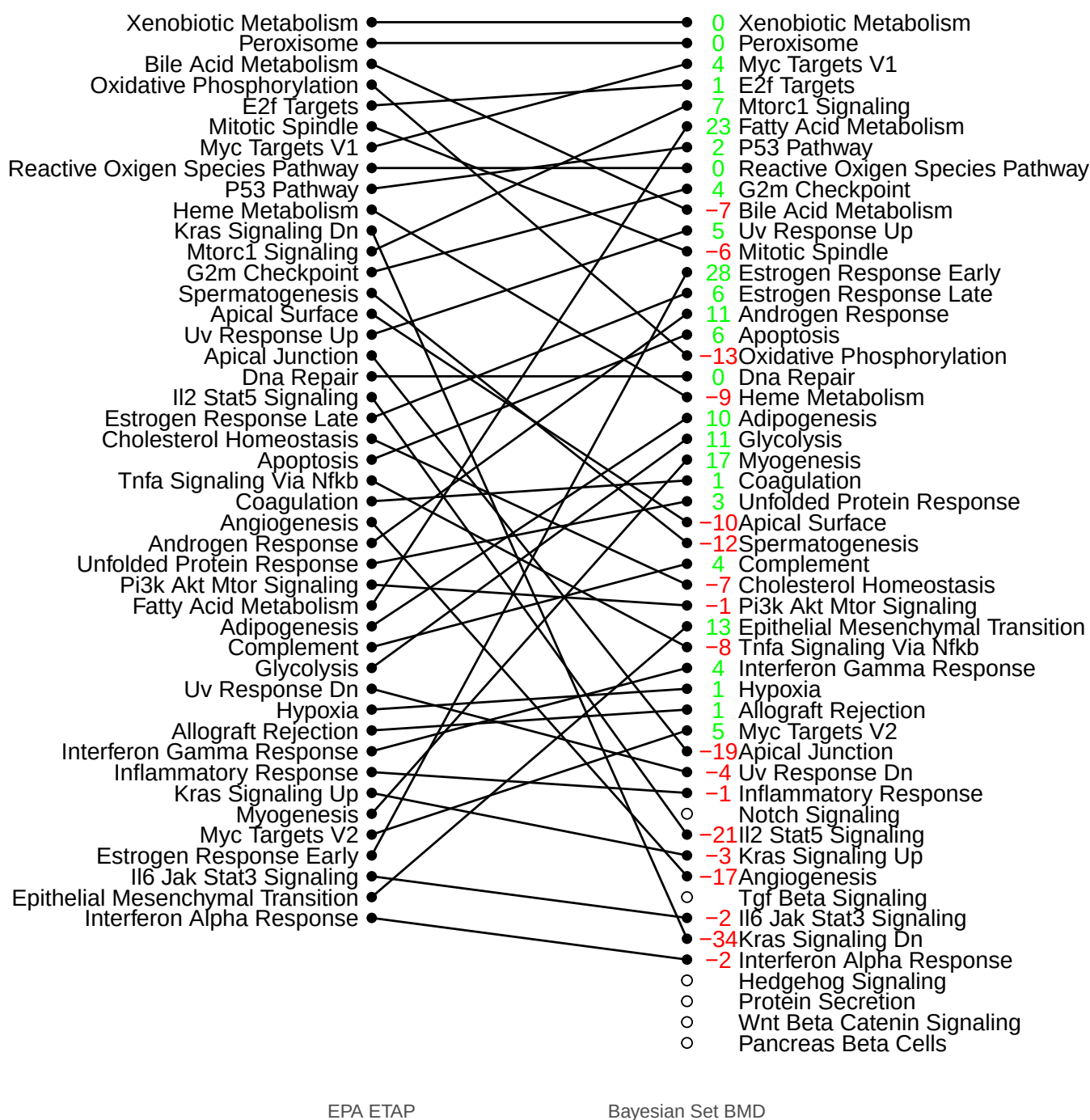


Figure 9: The hallmark sets are ordered by BMDL for methyleugenol exposure in the liver. Some pathways like interferon alpha response and apical junction have dramatic shifts in rank and appear to be unrelated to the potential organ damage pathways expected from methyleugenol. Numbers indicate difference in rank of hallmark set BMDLs between the two methods.

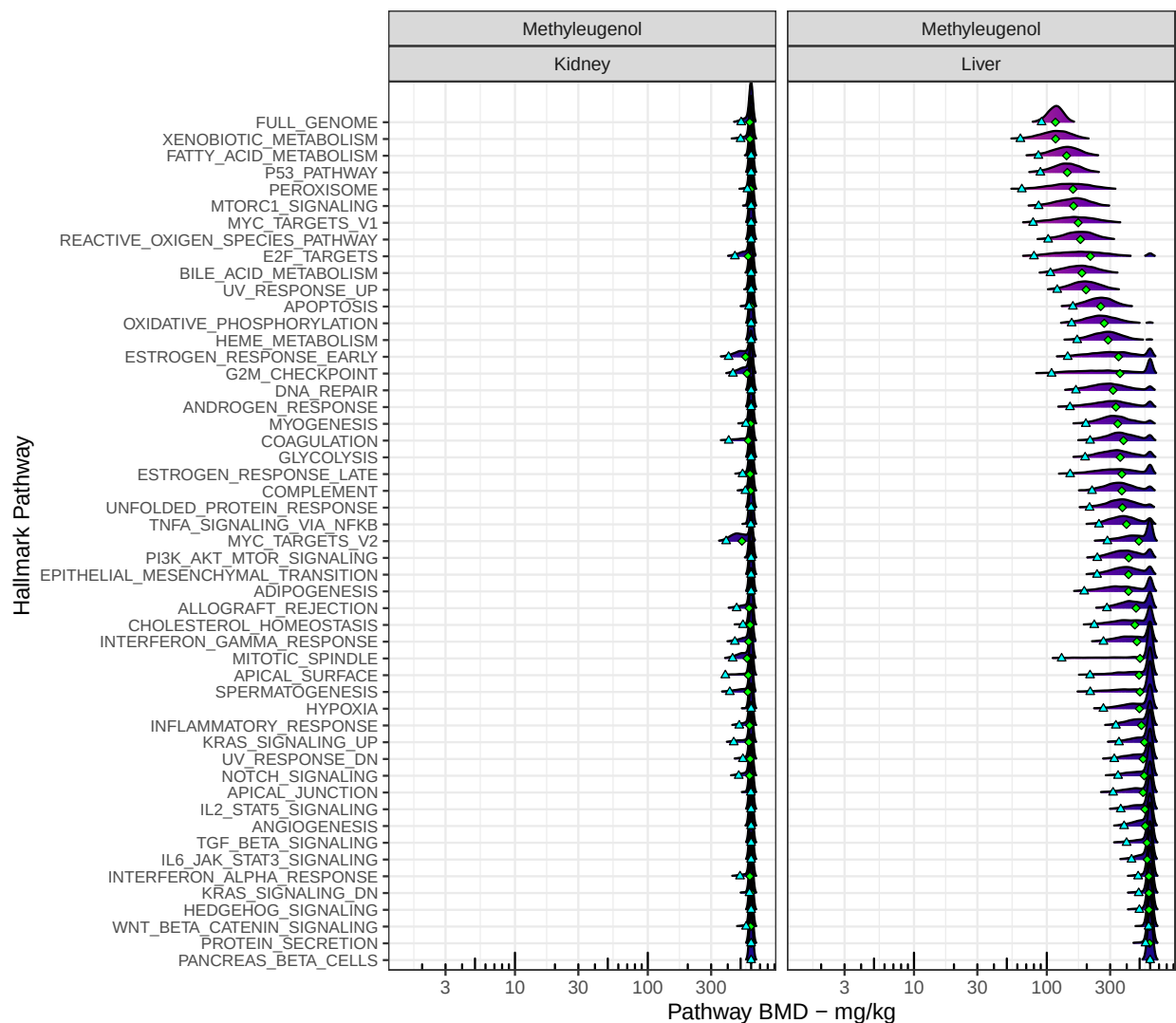


Figure 10: Methyleugenol BS-BMD simulations by each Hallmark pathway for kidney (left) and liver (right). The distribution of 5,000 posterior samples for the BS-BMD (X-Axis) are shown for each Hallmark gene set (Y-axis) as well as the median (diamonds), and 5th percentile (triangles). The maximum dose is returned as the BMD when the response is not significant, so the kidney shows no active pathways. The density is shown when the probability mass is at least 0.01.

8 BMD Tables: Results and comparisons

Comparisons between methods for their error against apical values provide a straightforward measure of accuracy, but can be misleading since the ETAP is tuned to fit apical values while our method is not adjusted or fit to any data. Compounding the issue is that the curated gene sets like the hallmark sets do not have well quantified, consistent mechanisms across all tissues. For example, our comparison to EPA methods shows that only a fraction of the genes in a set are active. Such results could suggest corrective or nonlinear mechanisms within the gene set that vary with tissue, but verification and simulation of these responses is beyond the scope of this work.

A notable distinction between the BS-BMD and ETAP results is the absence of a measurable response from number of chemical/tissue combinations with the BS-BMD approach. In particular ab-thujone which yielded no measurable response in either liver or kidney.

8.1 Scatterplot comparison

A BMDL comparison is visualized with a scatter plot in Figures 11 and 12. These plots summarize and compare the BS-BMD, the BMDExpress and apical benchmarks taken from supplementary table 5 of Gwinn et al. [2020]. The apical endpoints represent histopathological outcomes as the response rather than gene expression.

Figure 11 compares the lowest BMDL against the corresponding BS-BMDL estimate using Hallmark sets, along with each method compared to the apical estimate. This is an apples-to-oranges comparison because a change in gene expression may not be the cause of a pathological outcome, and the corresponding lowest BMDL may not even match the tissue of the pathology, which includes spleen and brain. For example, a compound like Ginseng was not found to cause any pathological outcomes, but responses in gene expression are detected in both the BS-BMD and EPA methods. Despite this issue, we find that the BS-BMDLs can be similar to the apical BMDs while providing meaningful pathway suggestions.

Figure 12 shows only comparisons to the apical endpoint, but with GO terms in addition to the Hallmark sets. Both the error and the visual pattern does not suggest that one method is overwhelmingly superior to the others. Extreme points are often found in pairs matched by gene set type, such as ETAP and BS-BMD values for Hallmark sets, implying that in those cases the two methods return similar estimates. As we argue in the main text and in the additional detailed analyses above, this is not necessarily meaningful because the GO terms or Hallmark sets that happen to supply the corresponding BMDs under the ETAP approach do not match the mechanistic understanding of how the chemical affects the body, and effects such as correlation were not accounted for. While we have not made any attempt to tune the parameters of our method to better match the apical outcomes, the ETAP protocol was designed and optimized with the GO sets to best match the apical data. As with the previous figure, this is an apples-to-oranges comparison since the apical tissues do not always match the tissues used to collect gene expression, so these results should be interpreted with caution.

8.2 Lowest Three Hallmark Gene Set BMDL

1	Acrylamide	Kidney	Bile Acid Metabolism	10.00	3.06
2	Acrylamide	Kidney	Mtorc1 Signaling	10.00	3.54
3	Acrylamide	Kidney	Tnfa Signaling Via Nfkb	10.00	3.77
4	Acrylamide	Liver	Tnfa Signaling Via Nfkb	10.00	8.14
5	Acrylamide	Liver	Wnt Beta Catenin Signaling	10.00	9.36
6	Acrylamide	Liver	Notch Signaling	10.00	9.41
7	BDCA	Kidney	Oxidative Phosphorylation	160.00	4.07
8	BDCA	Kidney	Inflammatory Response	160.00	6.38
9	BDCA	Kidney	Il2 Stat5 Signaling	160.00	7.42
10	BDCA	Liver	Fatty Acid Metabolism	129.88	90.69
11	BDCA	Liver	Peroxisome	160.00	116.51
12	BDCA	Liver	Oxidative Phosphorylation	160.00	133.07
13	BPAF	Kidney	Bile Acid Metabolism	278.98	119.96

14	BPAF	Kidney	Spermatogenesis	381.62	138.77
15	BPAF	Kidney	Xenobiotic Metabolism	369.83	197.13
16	BPAF	Liver	Epithelial Mesenchymal Transition	454.24	64.79
17	BPAF	Liver	Adipogenesis	189.44	73.65
18	BPAF	Liver	Uv Response Dn	435.65	100.20
19	Coumarin	Kidney	G2m Checkpoint	26.80	20.20
20	Coumarin	Kidney	E2f Targets	30.63	22.78
21	Coumarin	Kidney	Mitotic Spindle	30.62	22.93
22	Coumarin	Liver	Xenobiotic Metabolism	32.35	22.09
23	Coumarin	Liver	Myc Targets V2	81.13	23.41
24	Coumarin	Liver	Reactive Oxygen Species Pathway	54.67	27.99
25	DE71	Kidney	Inflammatory Response	465.63	2.84
26	DE71	Kidney	Apoptosis	264.27	3.43
27	DE71	Kidney	Uv Response Up	253.39	6.01
28	DE71	Liver	Xenobiotic Metabolism	2.89	1.15
29	DE71	Liver	Cholesterol Homeostasis	34.46	1.24
30	DE71	Liver	G2m Checkpoint	71.15	1.28
31	DEHP	Kidney	Fatty Acid Metabolism	177.46	100.74
32	DEHP	Kidney	Peroxisome	472.18	209.73
33	DEHP	Kidney	Oxidative Phosphorylation	1000.00	238.59
34	DEHP	Liver	Cholesterol Homeostasis	148.12	1.57
35	DEHP	Liver	Fatty Acid Metabolism	30.54	21.13
36	DEHP	Liver	Oxidative Phosphorylation	47.48	29.62
37	EE	Kidney	Bile Acid Metabolism	48.60	33.00
38	EE	Kidney	Hypoxia	48.60	37.93
39	EE	Kidney	Tnfa Signaling Via Nfkb	48.60	41.11
40	EE	Liver	Adipogenesis	4.40	2.35
41	EE	Liver	Cholesterol Homeostasis	5.75	3.36
42	EE	Liver	Mtorc1 Signaling	9.44	5.18
43	Fenofibrate	Kidney	Fatty Acid Metabolism	0.81	0.61
44	Fenofibrate	Kidney	Coagulation	2.07	1.11
45	Fenofibrate	Kidney	Estrogen Response Late	2.05	1.14
46	Fenofibrate	Liver	Fatty Acid Metabolism	0.92	0.68
47	Fenofibrate	Liver	Peroxisome	1.83	1.16
48	Fenofibrate	Liver	Oxidative Phosphorylation	2.43	1.48
49	Furan	Kidney	Coagulation	16.00	12.42
50	Furan	Kidney	Mitotic Spindle	16.00	13.54
51	Furan	Kidney	E2f Targets	16.00	14.70
52	Furan	Liver	E2f Targets	2.40	1.90
53	Furan	Liver	Mitotic Spindle	2.69	1.98
54	Furan	Liver	G2m Checkpoint	2.59	2.08
55	Ginseng	Kidney	Bile Acid Metabolism	5000.00	1468.14
56	Ginseng	Kidney	Peroxisome	5000.00	1835.01
57	Ginseng	Kidney	G2m Checkpoint	5000.00	2808.57
58	Ginseng	Liver	Apoptosis	5000.00	1347.51
59	Ginseng	Liver	Tnfa Signaling Via Nfkb	5000.00	2573.04
60	Ginseng	Liver	Reactive Oxygen Species Pathway	5000.00	4408.22
61	HCb	Kidney	Tnfa Signaling Via Nfkb	64.00	64.00
62	HCb	Kidney	Hypoxia	64.00	64.00
63	HCb	Kidney	Cholesterol Homeostasis	64.00	64.00
64	HCb	Liver	Xenobiotic Metabolism	8.95	5.67
65	HCb	Liver	Uv Response Up	15.25	8.92
66	HCb	Liver	Fatty Acid Metabolism	14.67	9.39
67	Methyleugenol	Kidney	Apical Surface	600.00	371.36

68	Methyleugenol	Kidney	Myc Targets V2	508.89	383.05
69	Methyleugenol	Kidney	Estrogen Response Early	579.58	404.32
70	Methyleugenol	Liver	Xenobiotic Metabolism	116.57	67.16
71	Methyleugenol	Liver	Peroxisome	149.64	67.60
72	Methyleugenol	Liver	Myc Targets V1	160.72	78.29
73	MilkThistleExtract	Kidney	Mitotic Spindle	1750.00	1505.23
74	MilkThistleExtract	Kidney	G2m Checkpoint	1750.00	1514.51
75	MilkThistleExtract	Kidney	E2f Targets	1750.00	1518.55
76	MilkThistleExtract	Liver	G2m Checkpoint	1750.00	1524.89
77	MilkThistleExtract	Liver	Tnfa Signaling Via Nfkb	1750.00	1564.89
78	MilkThistleExtract	Liver	Estrogen Response Early	1750.00	1584.27
79	PFOA	Kidney	Tnfa Signaling Via Nfkb	0.15	0.12
80	PFOA	Kidney	Hypoxia	0.22	0.16
81	PFOA	Kidney	Estrogen Response Late	0.26	0.17
82	PFOA	Liver	Tnfa Signaling Via Nfkb	0.13	0.11
83	PFOA	Liver	Hypoxia	0.14	0.11
84	PFOA	Liver	P53 Pathway	0.20	0.15
85	Pulegone	Kidney	Estrogen Response Late	60.19	36.28
86	Pulegone	Kidney	P53 Pathway	70.99	41.74
87	Pulegone	Kidney	Hypoxia	67.90	42.58
88	Pulegone	Liver	Myc Targets V2	85.98	11.98
89	Pulegone	Liver	Androgen Response	78.99	12.91
90	Pulegone	Liver	Myc Targets V1	72.10	19.16
91	TBBPA	Kidney	Coagulation	2000.00	1445.12
92	TBBPA	Kidney	E2f Targets	2000.00	1922.06
93	TBBPA	Kidney	Tnfa Signaling Via Nfkb	2000.00	2000.00
94	TBBPA	Liver	Cholesterol Homeostasis	2000.00	8.75
95	TBBPA	Liver	Myc Targets V2	1669.93	9.88
96	TBBPA	Liver	Hypoxia	190.95	9.89
97	TCAB	Kidney	G2m Checkpoint	9.53	0.82
98	TCAB	Kidney	E2f Targets	12.04	2.75
99	TCAB	Kidney	Mitotic Spindle	25.25	6.12
100	TCAB	Liver	Xenobiotic Metabolism	3.07	0.99
101	TCAB	Liver	Fatty Acid Metabolism	14.19	7.65
102	TCAB	Liver	Uv Response Up	17.32	9.86
103	TCPP	Kidney	Fatty Acid Metabolism	1429.50	917.49
104	TCPP	Kidney	Xenobiotic Metabolism	1720.80	1094.04
105	TCPP	Kidney	Coagulation	1729.72	1118.66
106	TCPP	Liver	Cholesterol Homeostasis	554.71	82.21
107	TCPP	Liver	Tnfa Signaling Via Nfkb	998.38	185.78
108	TCPP	Liver	Mtorc1 Signaling	899.64	212.66
109	Triclosan	Kidney	E2f Targets	378.94	269.24
110	Triclosan	Kidney	Complement	421.61	273.29
111	Triclosan	Kidney	Xenobiotic Metabolism	522.15	279.93
112	Triclosan	Liver	Cholesterol Homeostasis	131.59	3.45
113	Triclosan	Liver	Tnfa Signaling Via Nfkb	617.96	12.43
114	Triclosan	Liver	Fatty Acid Metabolism	29.70	13.83
115	abthujone	Kidney	Tnfa Signaling Via Nfkb	200.00	200.00
116	abthujone	Kidney	Hypoxia	200.00	200.00
117	abthujone	Kidney	Cholesterol Homeostasis	200.00	200.00
118	abthujone	Liver	Cholesterol Homeostasis	200.00	170.78
119	abthujone	Liver	Tnfa Signaling Via Nfkb	200.00	200.00
120	abthujone	Liver	Hypoxia	200.00	200.00

Table 3: The three lowest BMDL values are reported for each tissue and chemical. When all hallmark sets show equal BMD values, three are chosen at random.

8.3 BMD Comparison, Apical Endpoints

Table 4: Comparison of apical endpoint BMD to our BS-BMD. Apical endpoints taken from supplemental table 5 of Gwinn et al. [2020]. BS-BMD levels are taken as the minimum across both tissues. NA values for the apical endpoint implies that there was no significant change in pathology. MDT for our method represents maximum dose tested and is equivalent to an NA for the apical data.

	Chemical	Apical (Histopathologic) Endpoint	BMD	BMDL	Tissue	Hallmark	BSBMD	BSEMDL
1	Acrylamide	Peripheral nerve (sciatic) axon degeneration	0.61	0.43	Kidney	Tnfa Signaling Via Nfkb	MDT	3.77
2	Acrylamide	Preputial gland duct ectasia	1.46	0.74	Kidney	Hypoxia	MDT	4.61
3	Acrylamide	Eye retina degeneration	2.04	1.34	Kidney	Cholesterol Homeostasis	MDT	10.00
4	BDCa	Bone marrow angiectasis	2.30	1.87	Liver	Fatty Acid Metabolism	129.88	90.69
5	BDCa	Bone marrow hyperplasia	20.03	6.49	Kidney	Tnfa Signaling Via Nfkb	MDT	MDT
6	BDCa	Liver eosinophilic focus	26.80	13.66	Kidney	Hypoxia	MDT	126.12
7	Coumarin	Liver necrosis	5.85	5.85	Kidney	G2m Checkpoint	26.80	20.20
8	Coumarin	Forestomach ulcer	7.96	6.14	Kidney	Mitotic Spindle	30.62	22.93
9	Coumarin	Liver fibrosis	14.40	7.40	Kidney	E2f Targets	30.63	22.78
10	DE71	Liver hepatocyte hypertrophy	0.15	0.11	Liver	Xenobiotic Metabolism	2.89	1.15
11	DEHP	Testes degeneration seminiferous tubules	183.50	155.34	Liver	Fatty Acid Metabolism	30.54	21.13
12	DEHP	Anterior pituitary cell hypertrophy	324.05	262.94	Liver	Oxidative Phosphorylation	47.48	29.62
13	EE	Mammary gland alveolar hyperplasia	0.00	0.00	Liver	Adipogenesis	4.40	2.35
14	Fenofibrate	Degeneration, granular, eosinophilic, hepatocyte	2.69	NA	Kidney	Fatty Acid Metabolism	0.81	0.61
15	Furan	Liver cholangiofibrosis	0.10	0.09	Liver	E2f Targets	2.40	1.90
16	Furan	Liver cytoplasmic vacuolization	0.18	0.14	Liver	G2m Checkpoint	2.59	2.08
17	Furan	Liver bile duct biliary tract hyperplasia	0.61	0.31	Liver	Mitotic Spindle	2.69	1.98
18	Ginseng	NA	NA	NA	Kidney	Tnfa Signaling Via Nfkb	MDT	MDT
19	HCB	Peribiliary lymphocytosis	1.17	0.33	Liver	Xenobiotic Metabolism	8.95	5.67
20	HCB	Chronic nephrosis (severe)	0.59	0.35	Liver	Fatty Acid Metabolism	14.67	9.39
21	HCB	Centrilobular basophilic chromogenesis (slight)	0.62	0.40	Liver	Uv Response Up	15.25	8.92
22	Methyleugenol	Liver eosinophilic focus	13.14	10.01	Liver	Xenobiotic Metabolism	116.57	67.16
23	Methyleugenol	Liver cystic degeneration	16.55	13.77	Liver	Fatty Acid Metabolism	138.50	88.80
24	Methyleugenol	Glandular stomach atrophy	16.63	14.17	Liver	P53 Pathway	141.49	91.80
25	PFOA	NA	NA	NA	Liver	Tnfa Signaling Via Nfkb	0.13	0.11
26	Pulegone	Liver hepatocyte cellular alteration diffuse	16.58	7.52	Liver	Mtorc1 Signaling	52.19	33.33
27	Pulegone	Liver bile duct hyperplasia	13.67	10.66	Kidney	Estrogen Response Late	60.19	36.28
28	Pulegone	Nose olfactory epithelium degeneration	9.47	14.02	Liver	Xenobiotic Metabolism	64.01	39.57
29	TCAB	Forestomach epithelium hyperplasia	2.86	2.13	Liver	Xenobiotic Metabolism	3.07	0.99
30	TCAB	Lung alveolar epithelium hyperplasia bronchiolar	7.71	5.38	Kidney	G2m Checkpoint	9.53	0.82
31	TCAB	Liver diffuse fatty change	9.30	7.17	Kidney	E2f Targets	12.04	2.75
32	TCPP	NA	NA	NA	Liver	Cholesterol Homeostasis	554.71	82.21
33	abthujone	Kidney mineralization	3.17	2.29	Kidney	Tnfa Signaling Via Nfkb	MDT	MDT
34	abthujone	Spleen pigmentation	10.45	4.67	Kidney	Hypoxia	MDT	MDT

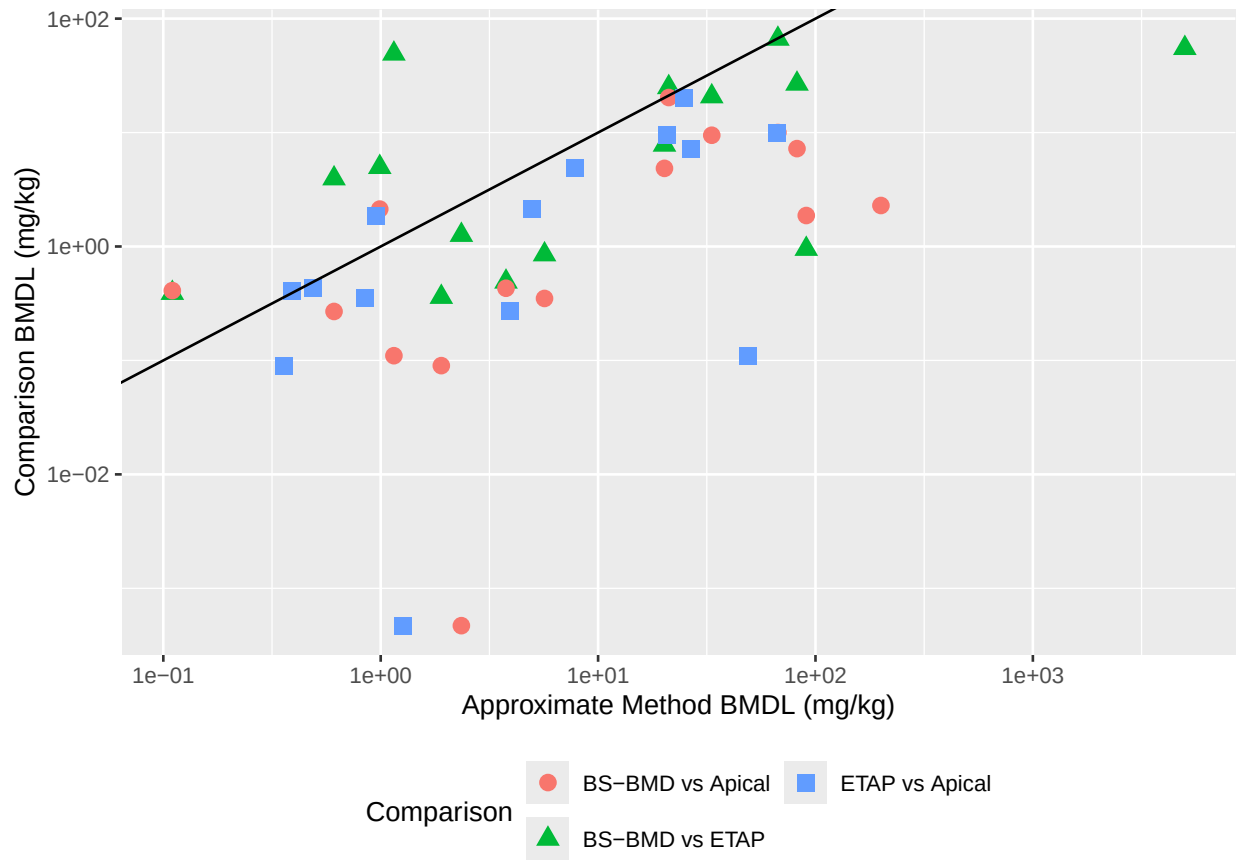


Figure 11: Three-way comparison of BMDL estimates from the EPA methodology and apical study to our BS-BMD method using a scatter plot and paired results using the Hallmark gene sets. BS-BMD tends to return larger values than the apical study and are comparable or larger than the EPA method. The points correspond to the lowest BMDL values reported for the EPA or apical study, if present, across all tissues.

8.4 BMD Comparison with GO terms, Lowest BMDL

This section contains tables and a plot comparing across all methods and gene sets. All apical results came from supplemental table 5 from Gwinn et al. [2020]. These comparisons suggest that one method is not uniformly superior to another when using hallmark sets. The ETAP method tends to better match the apical endpoint BMD when using GO terms, but in some cases no BMD is reported for the ETAP method while BS-BMD has a reasonably good estimate.

Table 5: Comparison of methods using a log ratio between the method and apical values, $\log(\text{BMDL} / \text{apical BMDL})$. Provides an estimated error in terms of orders of magnitude or fold change difference, with 0 implying no error. The practical difference is minimal between methods. Ginseng removed since there is no apical endpoint.

	Chemical	BS-BMDL	ETAP BMDL	BS-BMDL GO	ETAP BMDL GO
1	Acrylamide	2.2	0.13	-0.47	-0.43
2	BDCA	3.9	-0.68	-0.022	-0.69
3	Coumarin	1.4	0.48	0.92	-1.1
4	DE71	2.3	6.1	1.4	1.8
5	DEHP	0.04	0.21	-2.3	-0.37
6	EE	1.6	0.99	-0.67	-2.8
7	Fenofibrate	0.82	2.7	0.64	1
8	Furan	3	1.4	2.1	1.2
9	HCB	2.8	0.89	-1.8	-2.3
10	Methyleugenol	1.9	1.9	0.29	1.2
11	PFOA	-1.3	-0.05	-1.5	-3.5
12	Pulegone	1.3	0.78	-1.3	0.54
13	TCAB	-0.77	0.85	-1.5	-0.0038
14	TCPP	2.4	1.3	-0.4	0.13
	Mean	1.54	1.21	-0.33	-0.38

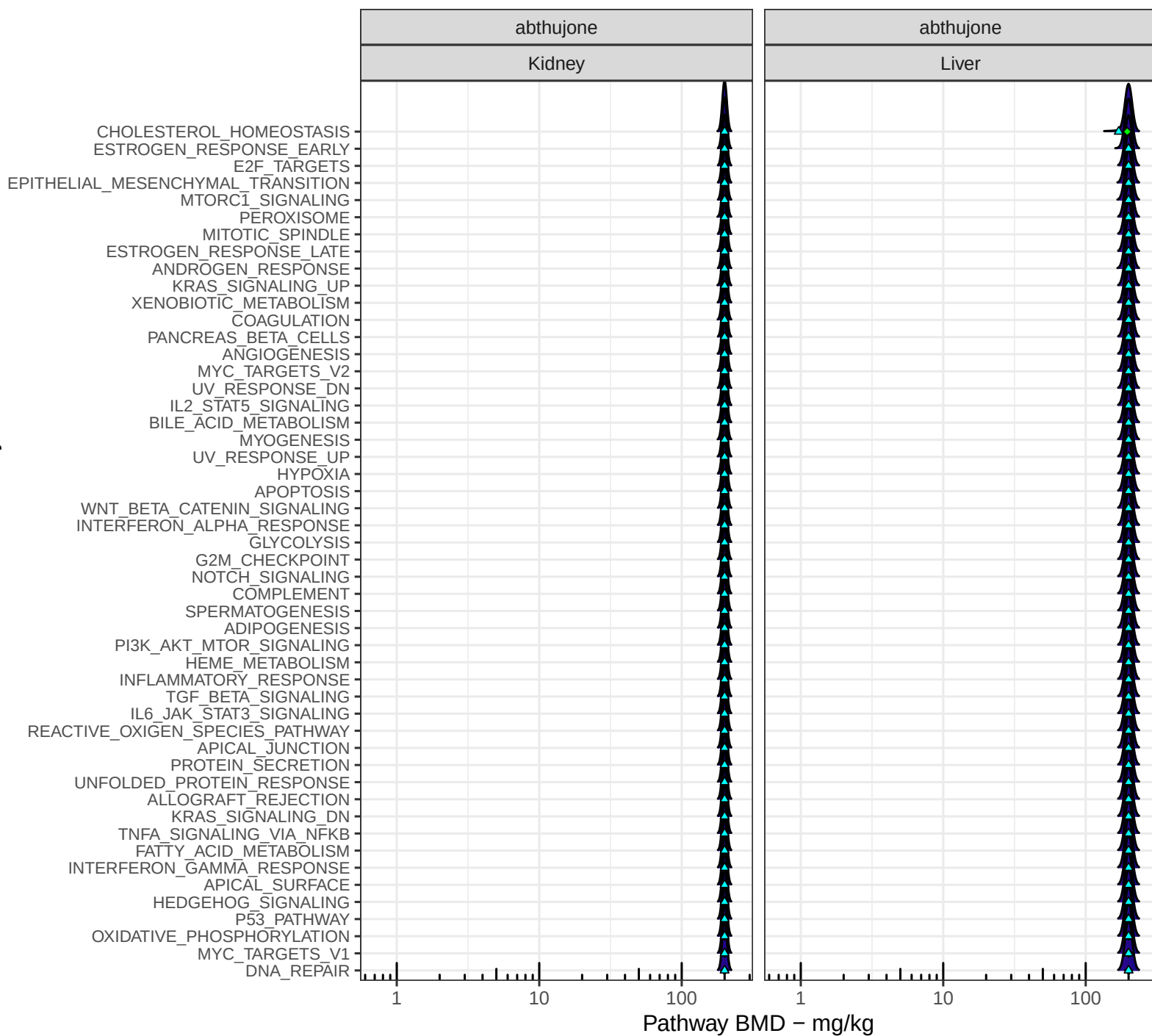
Table 6: Comparison of apical endpoint BMDL to BSBMD and ETAP methods using Hallmark sets or GO terms. Ap BMDL refers to the apical BMDL taken from the most recent EPA ETAP report and includes values from Table 5 of Gwinn et al. [2020], the methods with GO refer to the lowest discovered BMDL with GO sets, and the remaining columns refer to the lowest BMDL found for Hallmark sets. NA implies no finite value is available.

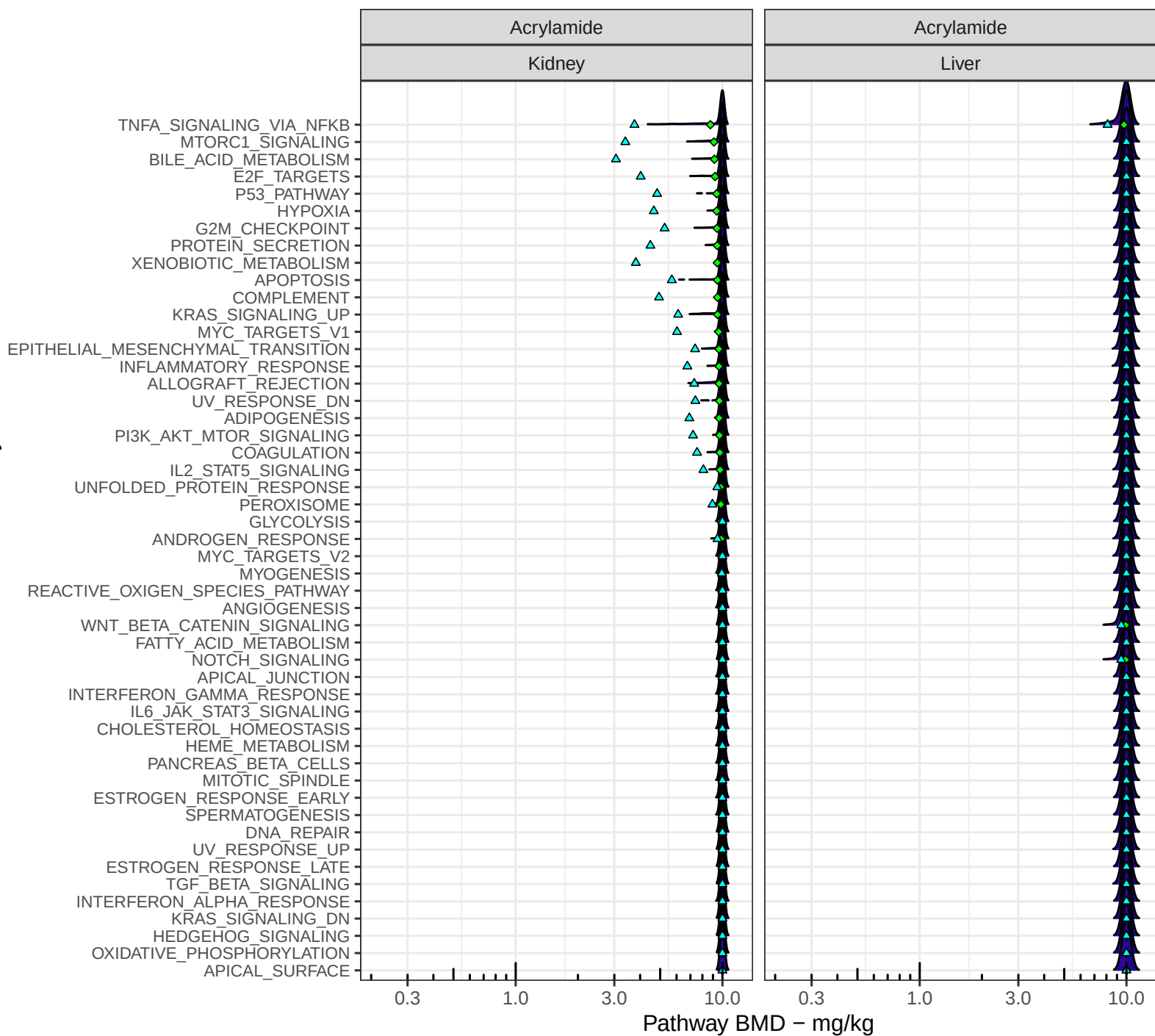
	Chemical	Ap.BMDL	BS-BMDL	ETAP BMDL	BS-BMDL GO	ETAP BMDL GO
1	Acrylamide	0.43	3.77	0.49	0.27	0.28
2	BDCA	1.87	90.69	0.95	1.83	0.94
3	Coumarin	4.85	20.20	7.80	12.14	1.57
4	DE71	0.11	1.15	49.15	0.43	0.66
5	DEHP	20.30	21.13	24.95	1.96	14.04
6	EE	0.00047	0.00235	0.00126	0.00024	2.9e-05
7	Fenofibrate	0.27	0.61	3.95	0.51	0.76
8	Furan	0.09	1.90	0.36	0.75	0.31
9	Ginseng	NA	5000.00	55.04	13.61	46.81
10	HCB	0.35	5.67	0.85	0.06	0.04
11	Methyleugenol	10.01	67.16	66.38	13.44	34.20
12	PFOA	0.41	0.11	0.39	0.09	0.01
13	Pulegone	9.47	33.33	20.73	2.63	16.27
14	TCAB	2.13	0.99	4.97	0.48	2.12
15	TCPP	7.23	82.21	26.77	4.86	8.22

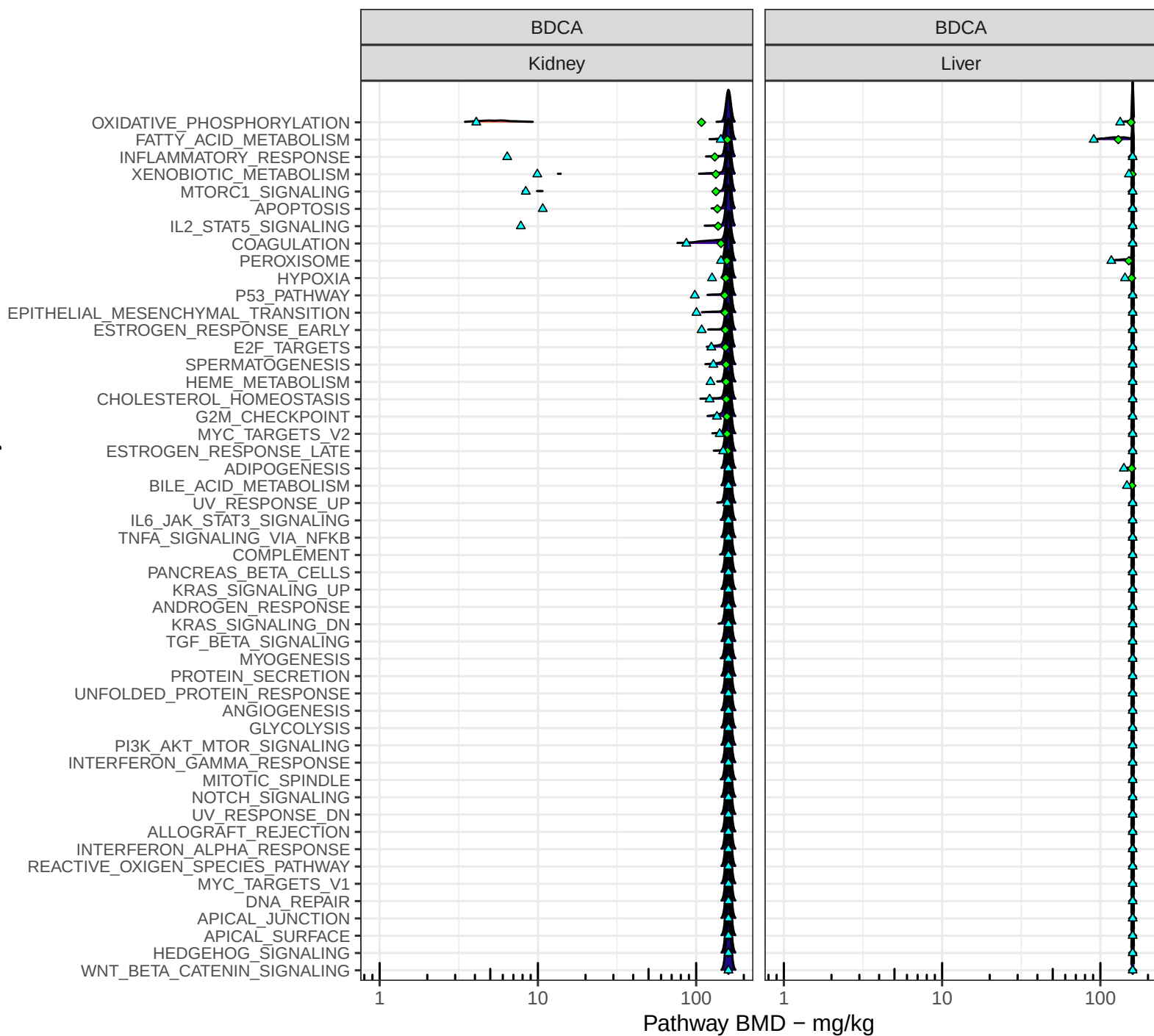
9 BMD Plots

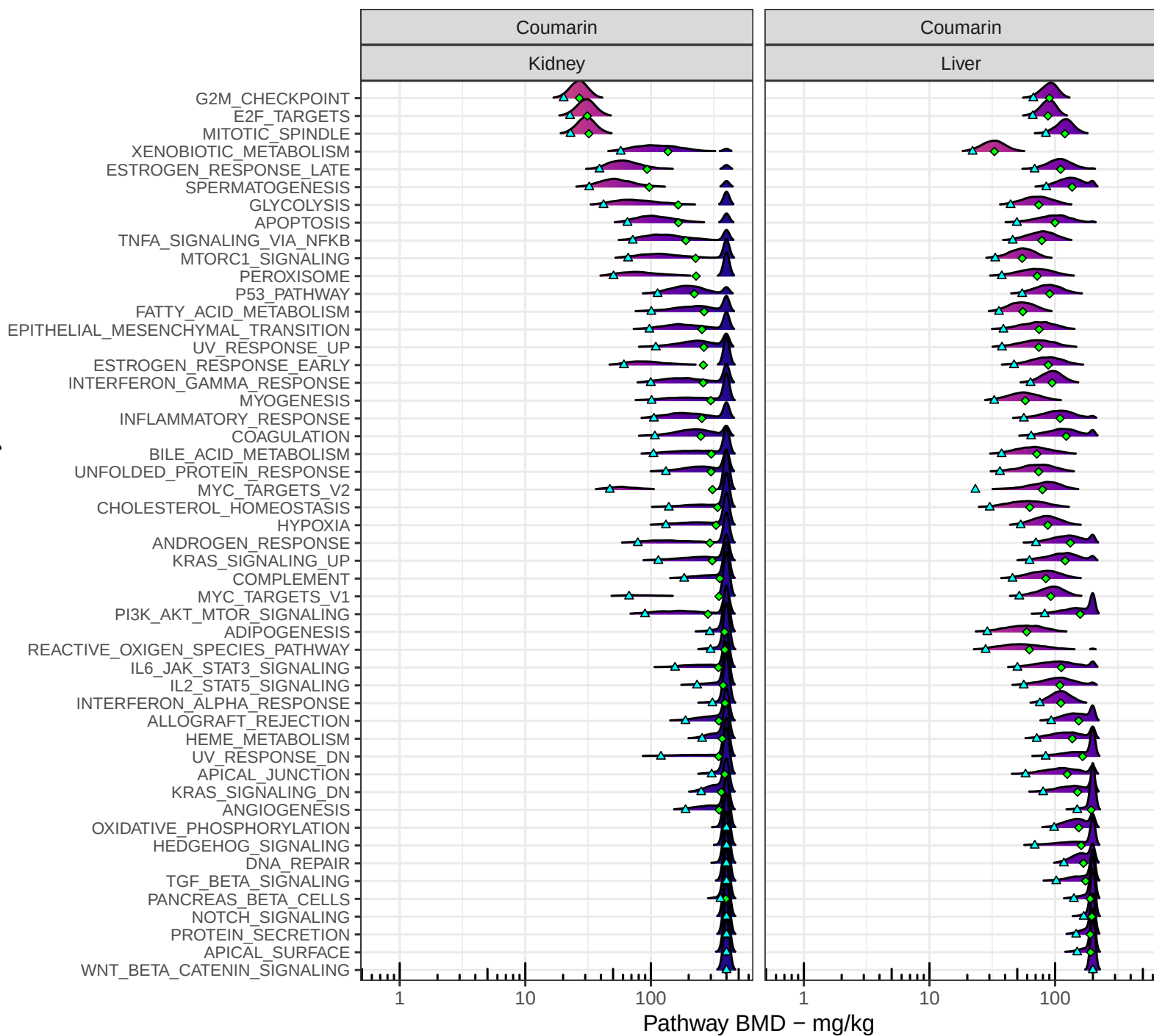
9.1 BMD posterior plots, All Tissues

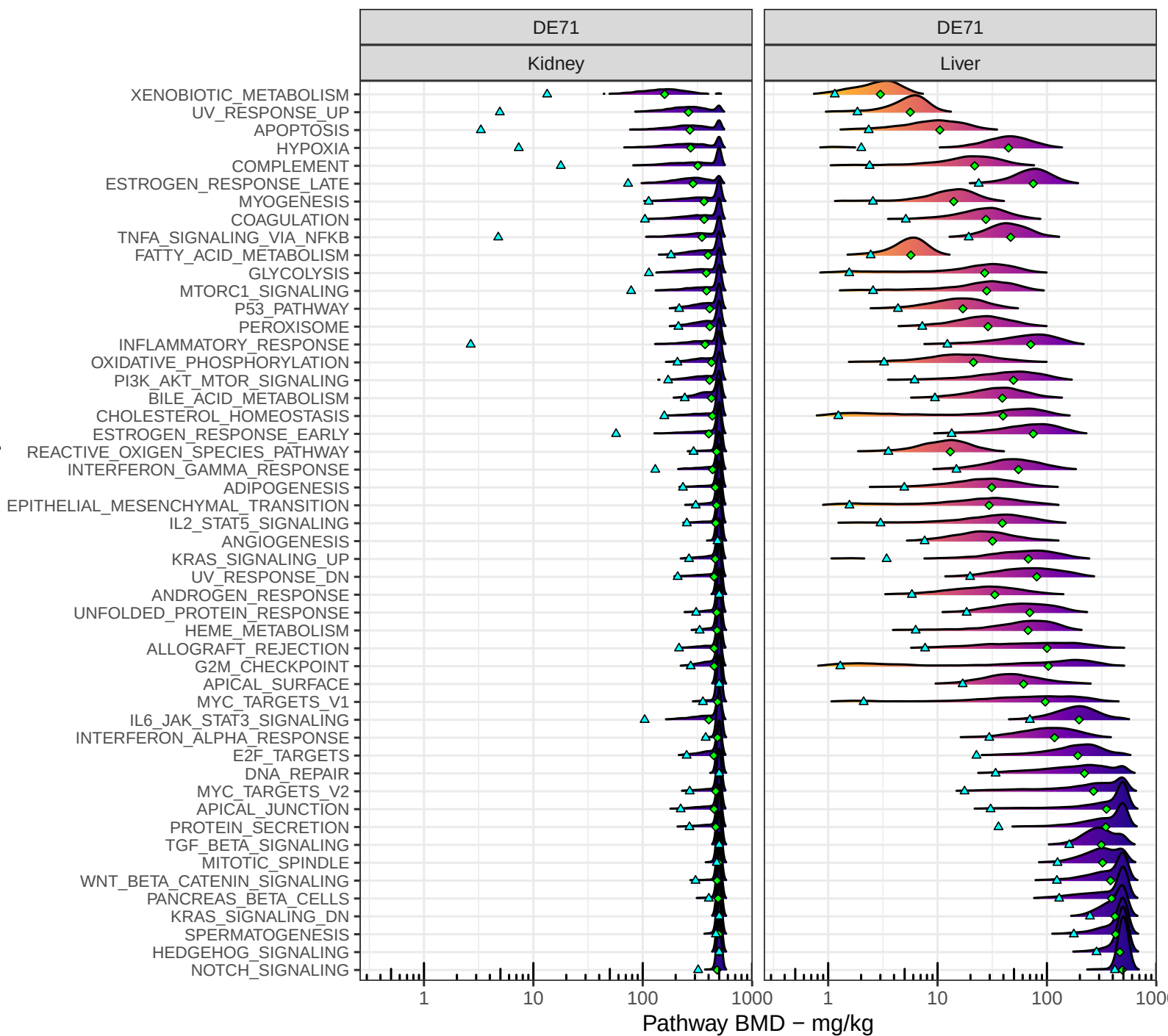
The posterior density plots below only show regions with mass above 0.01.

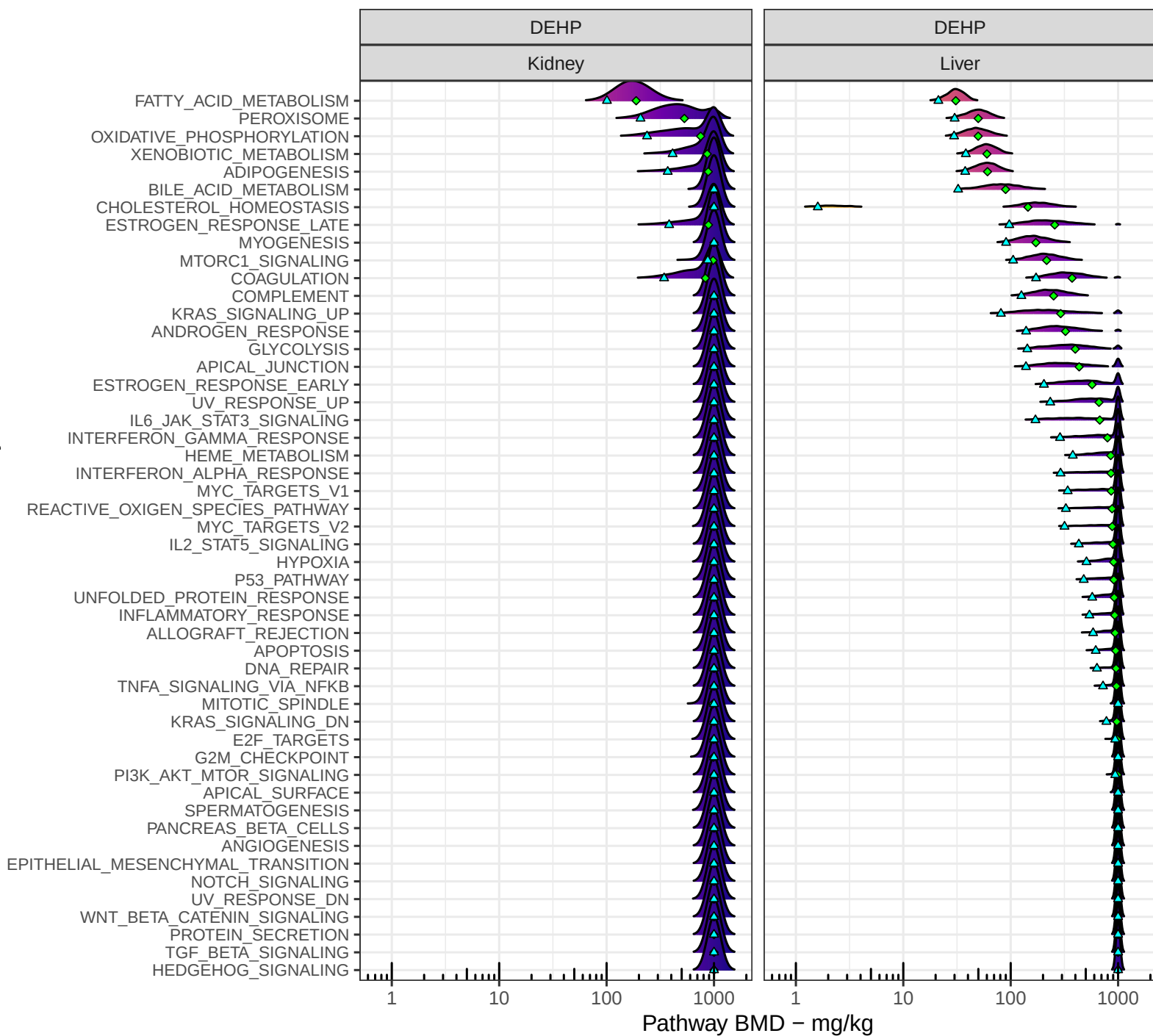


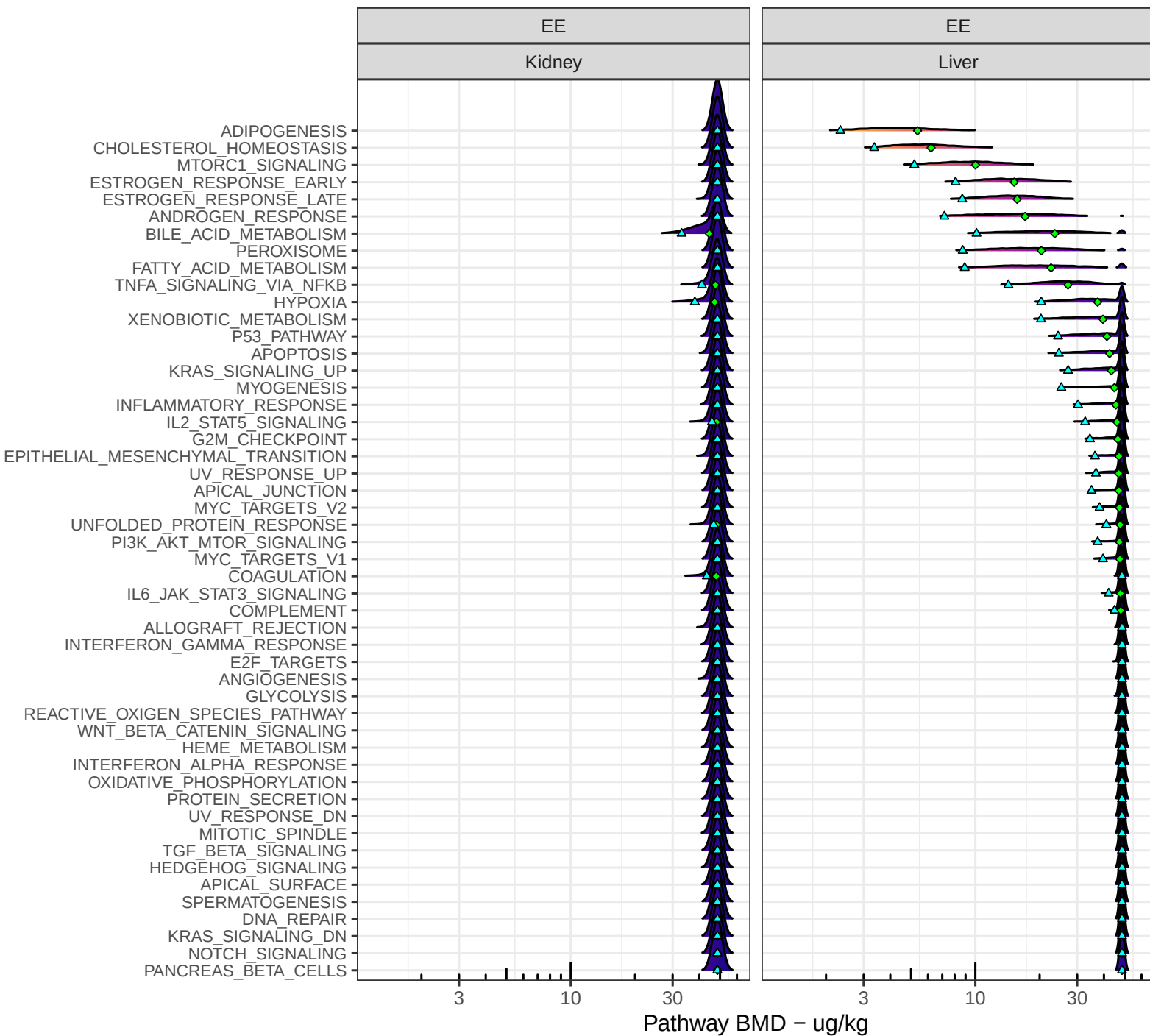


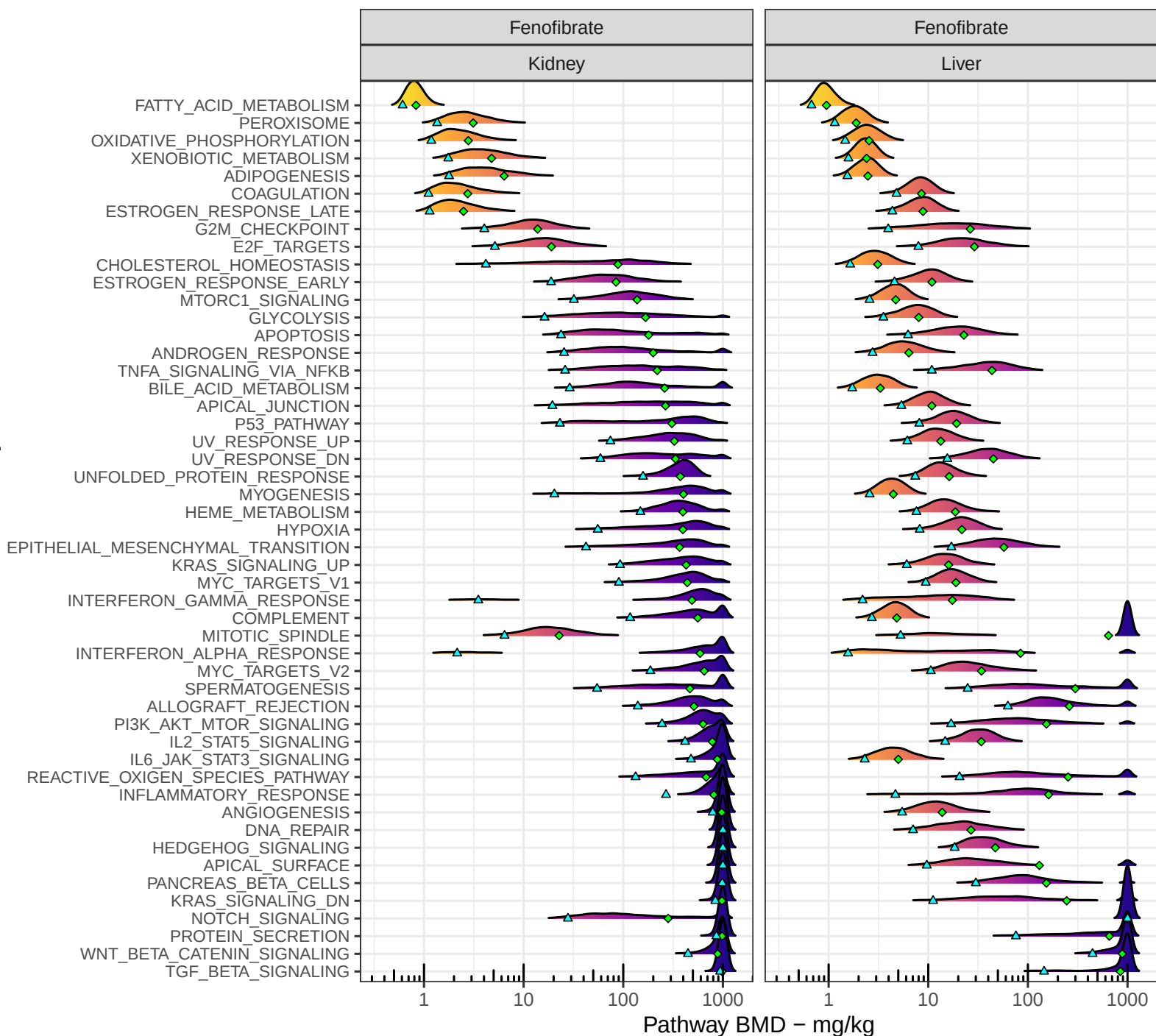


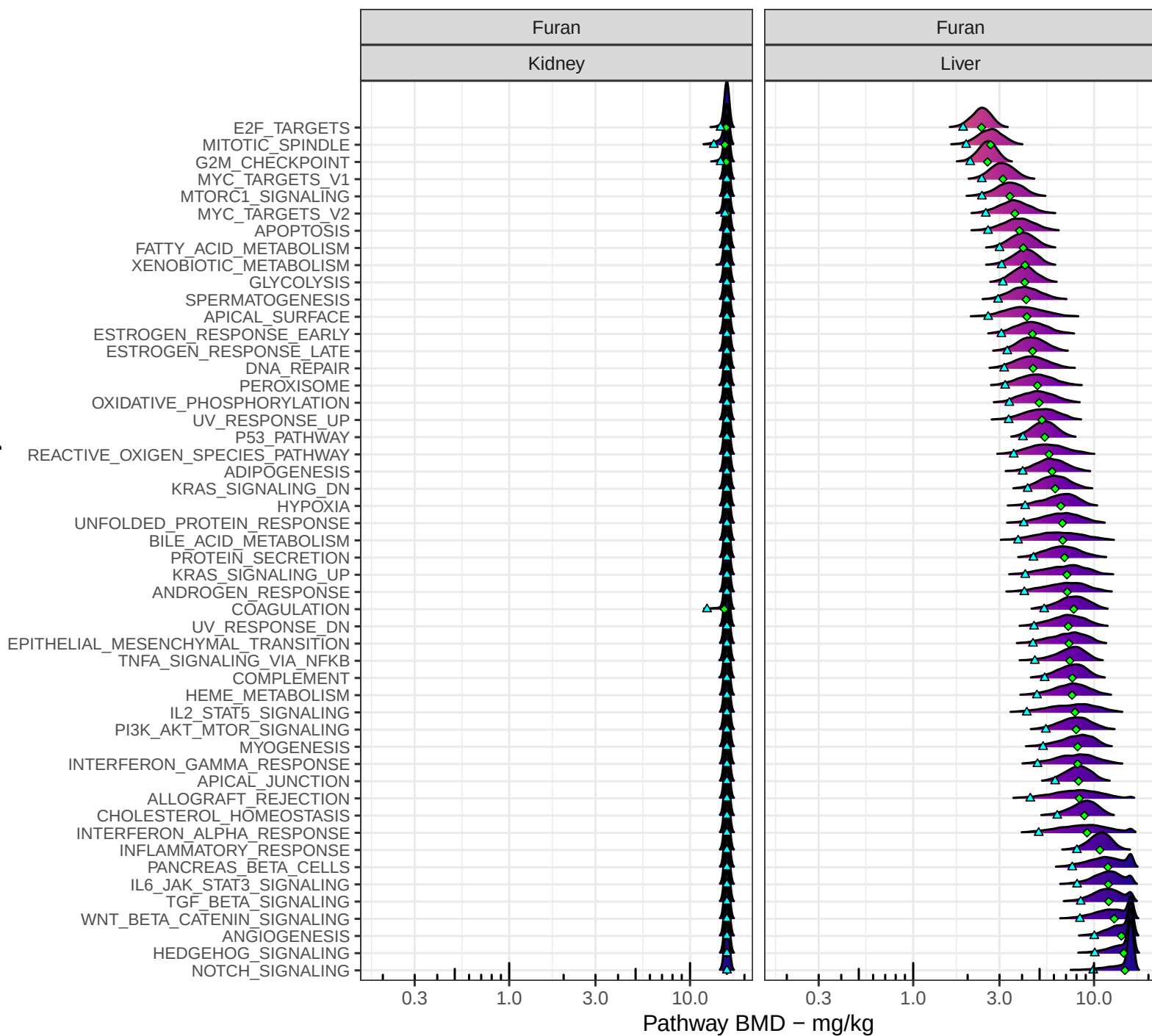


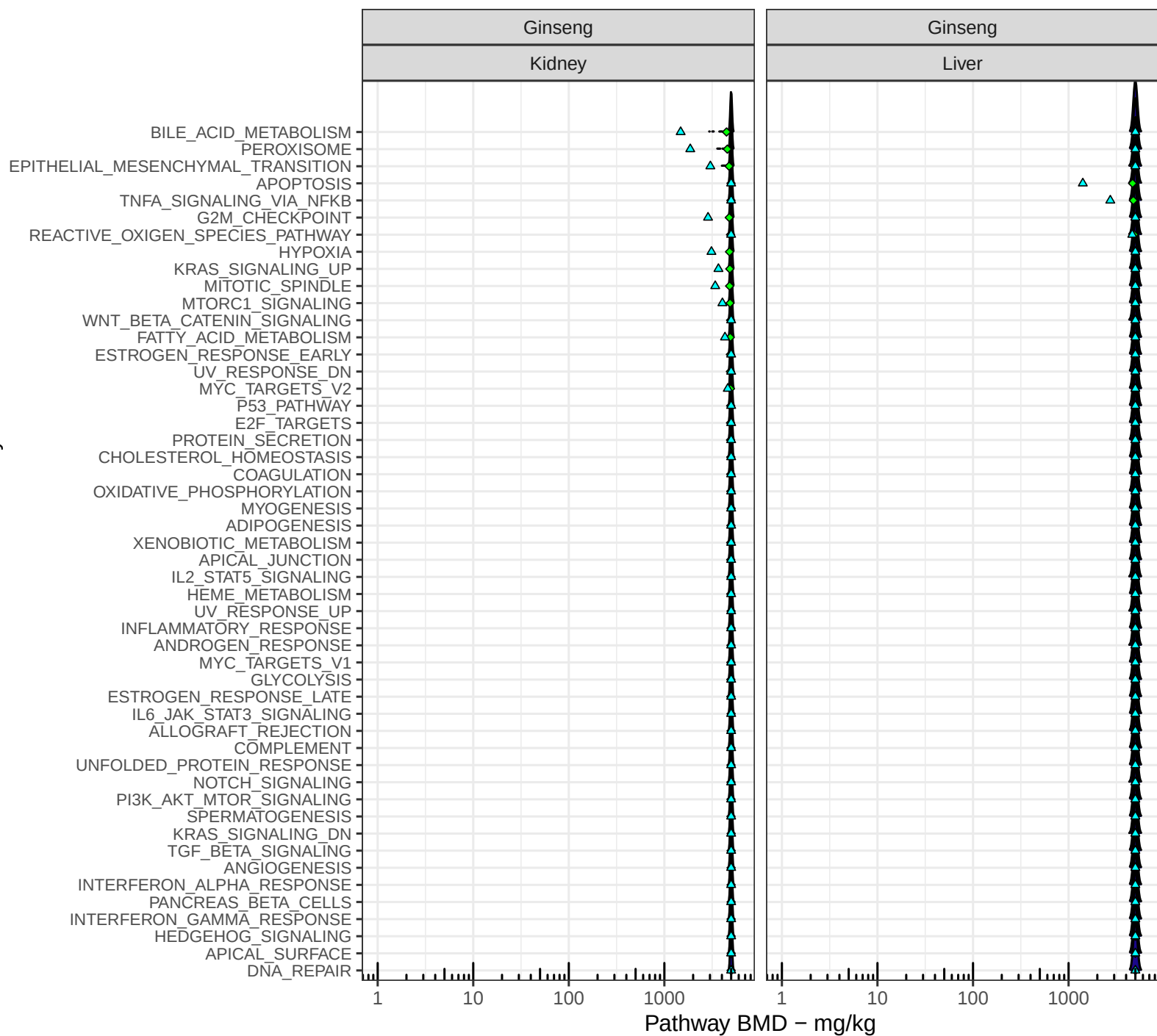


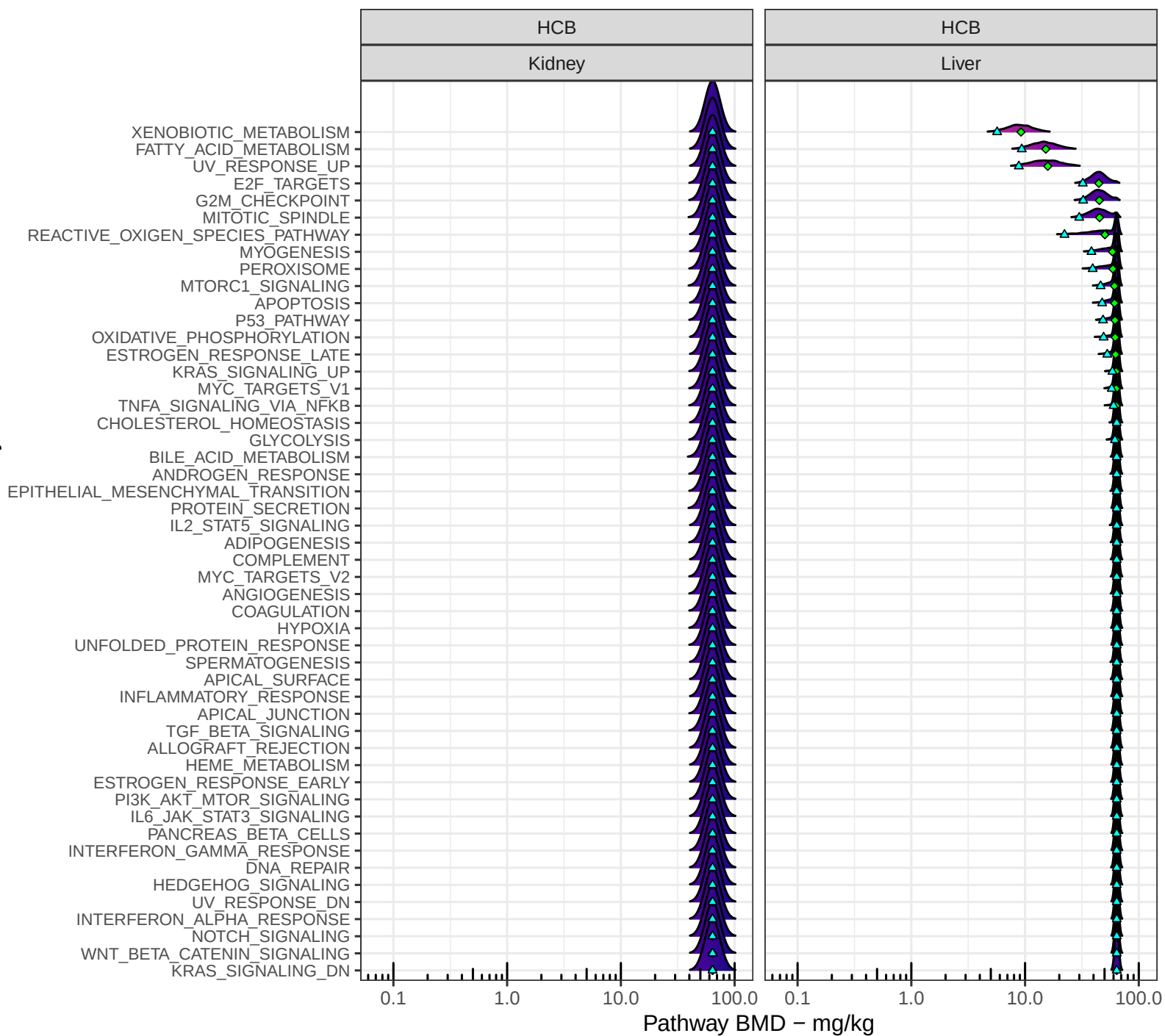


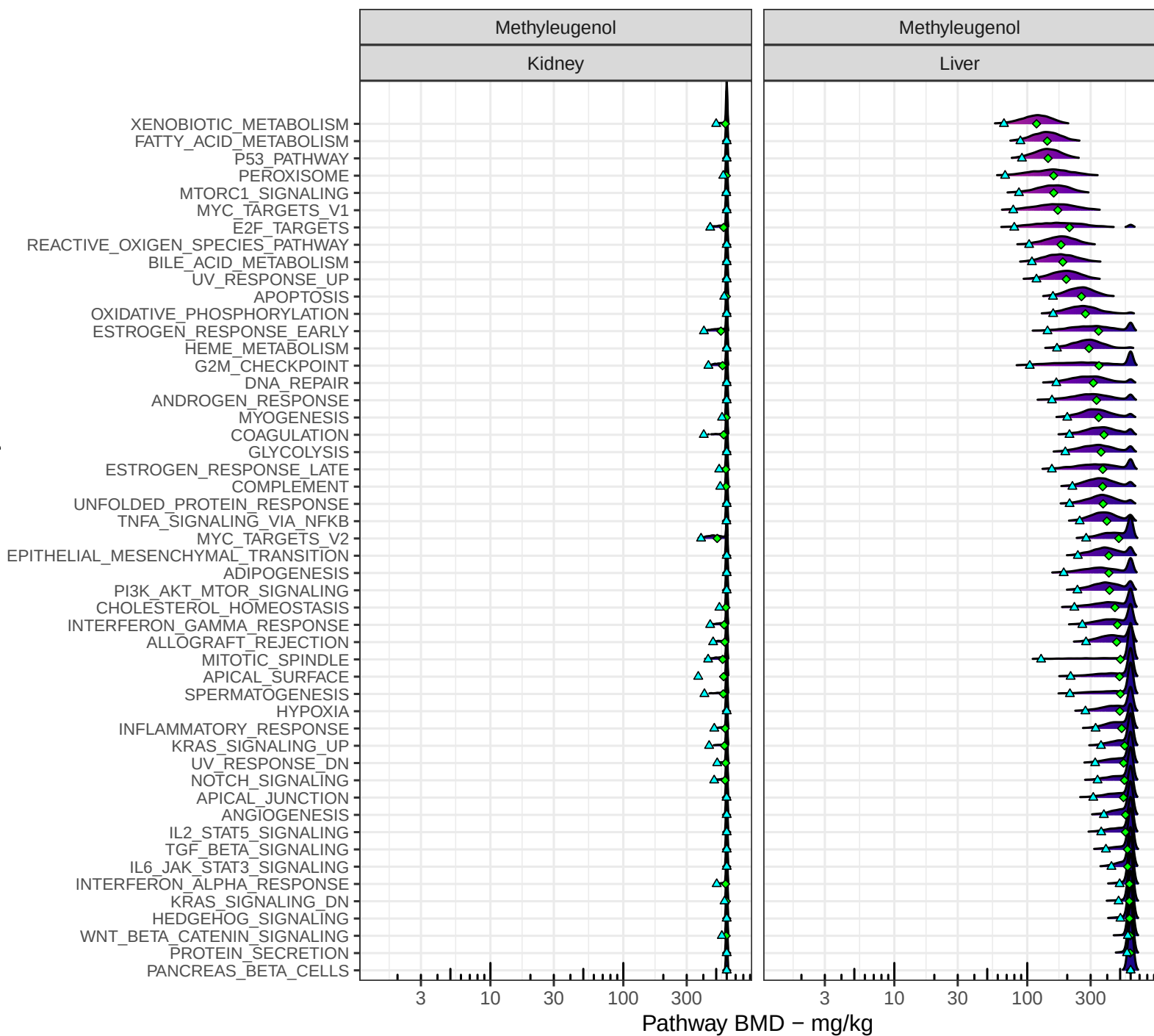


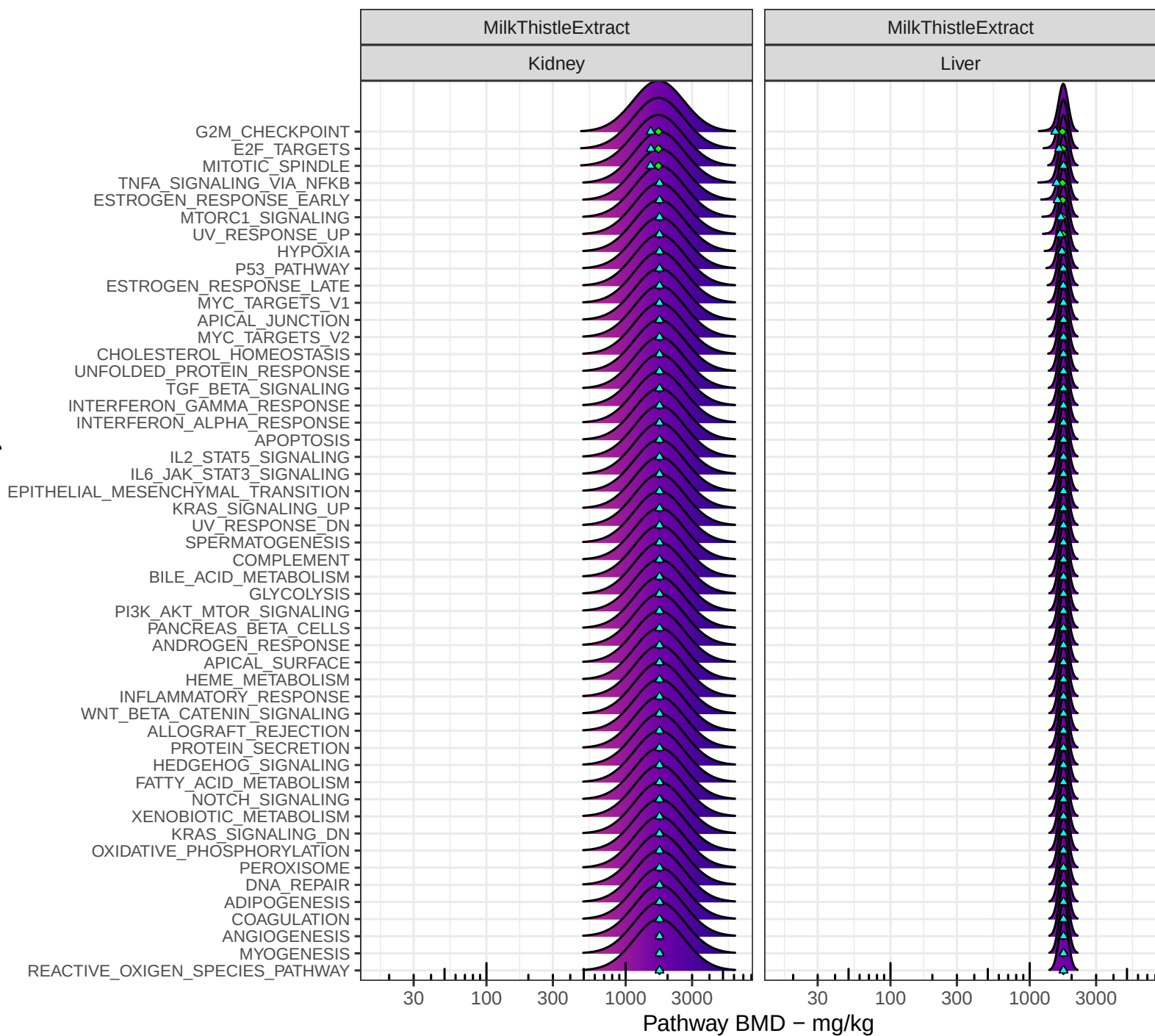


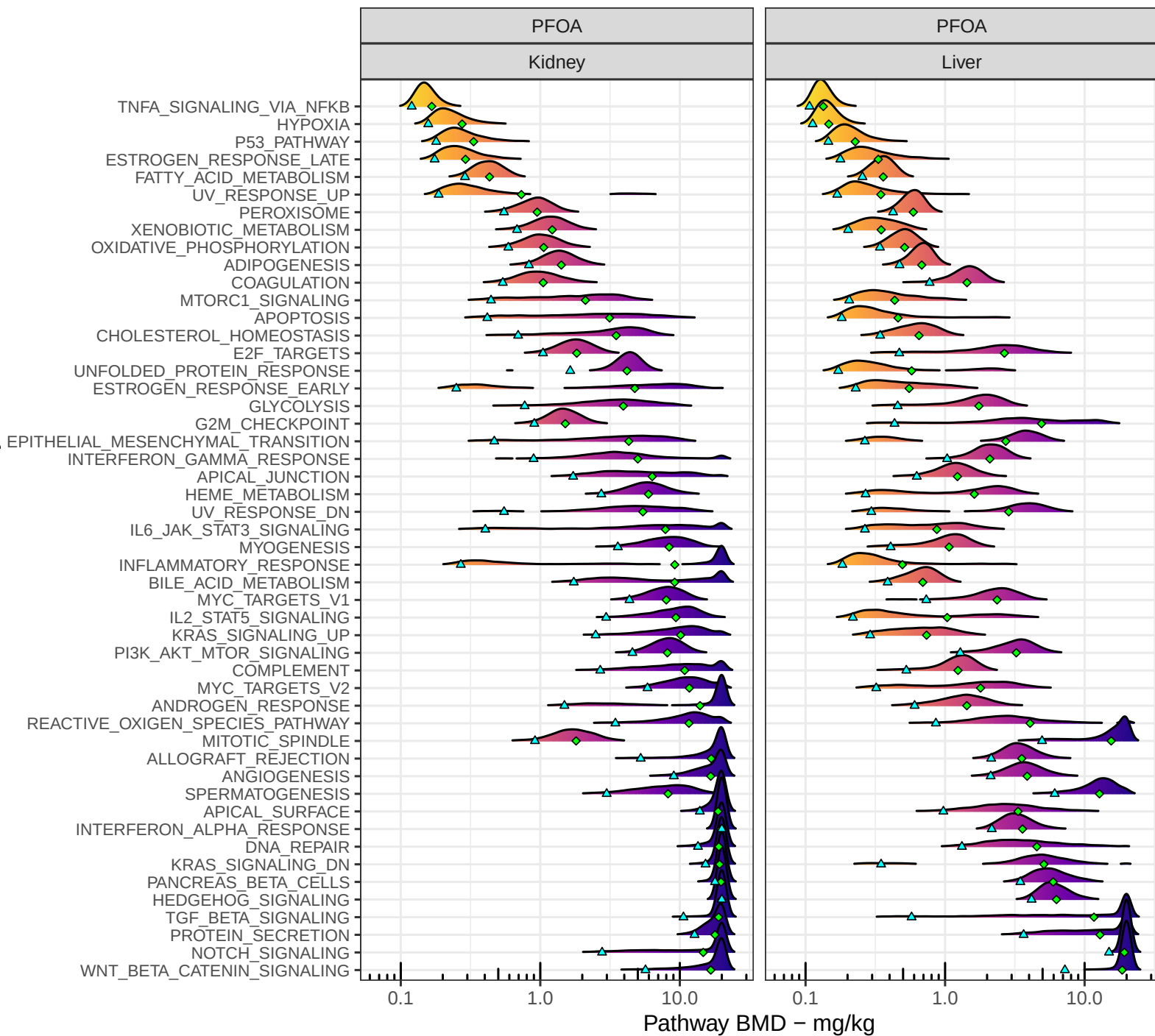


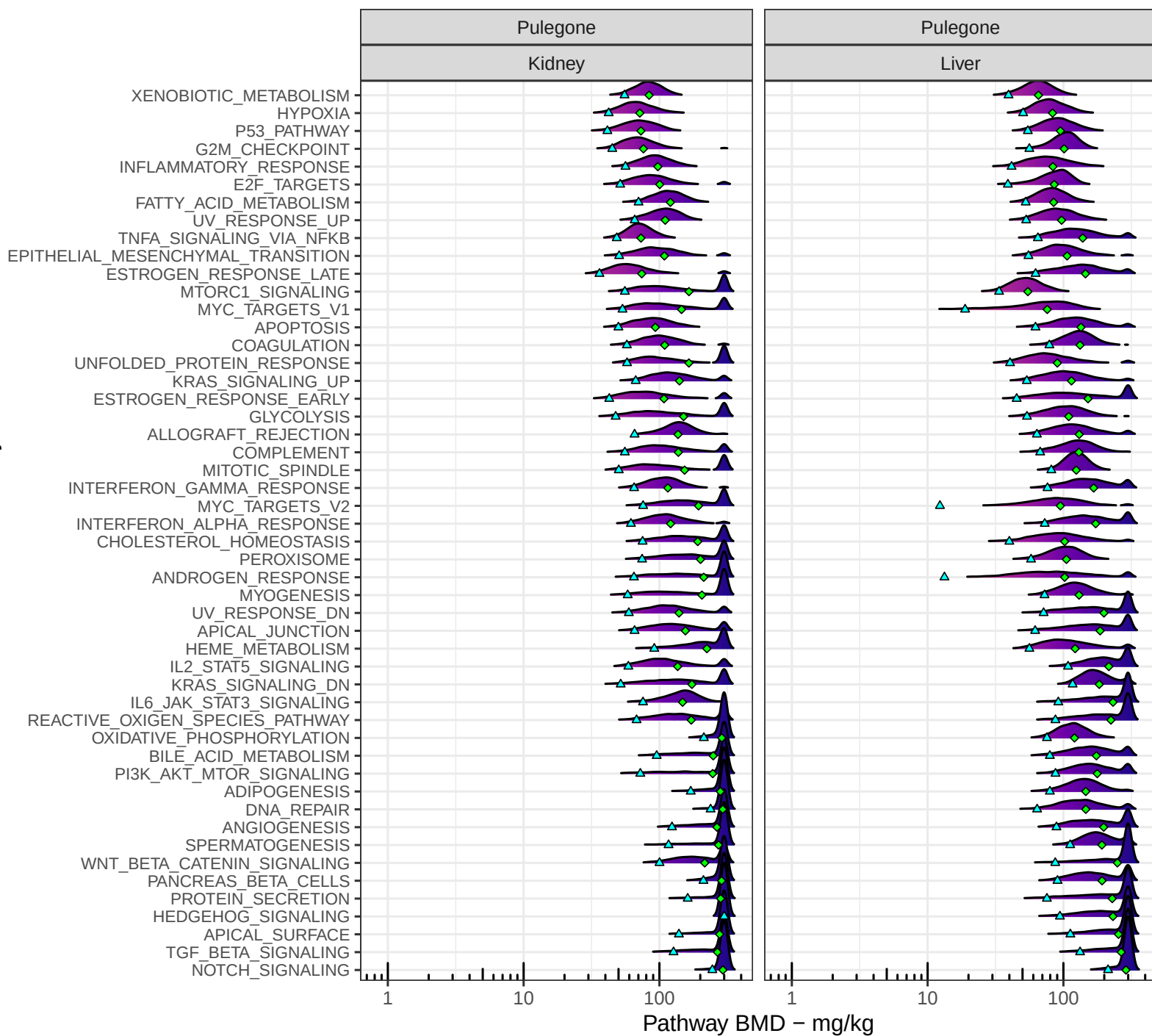


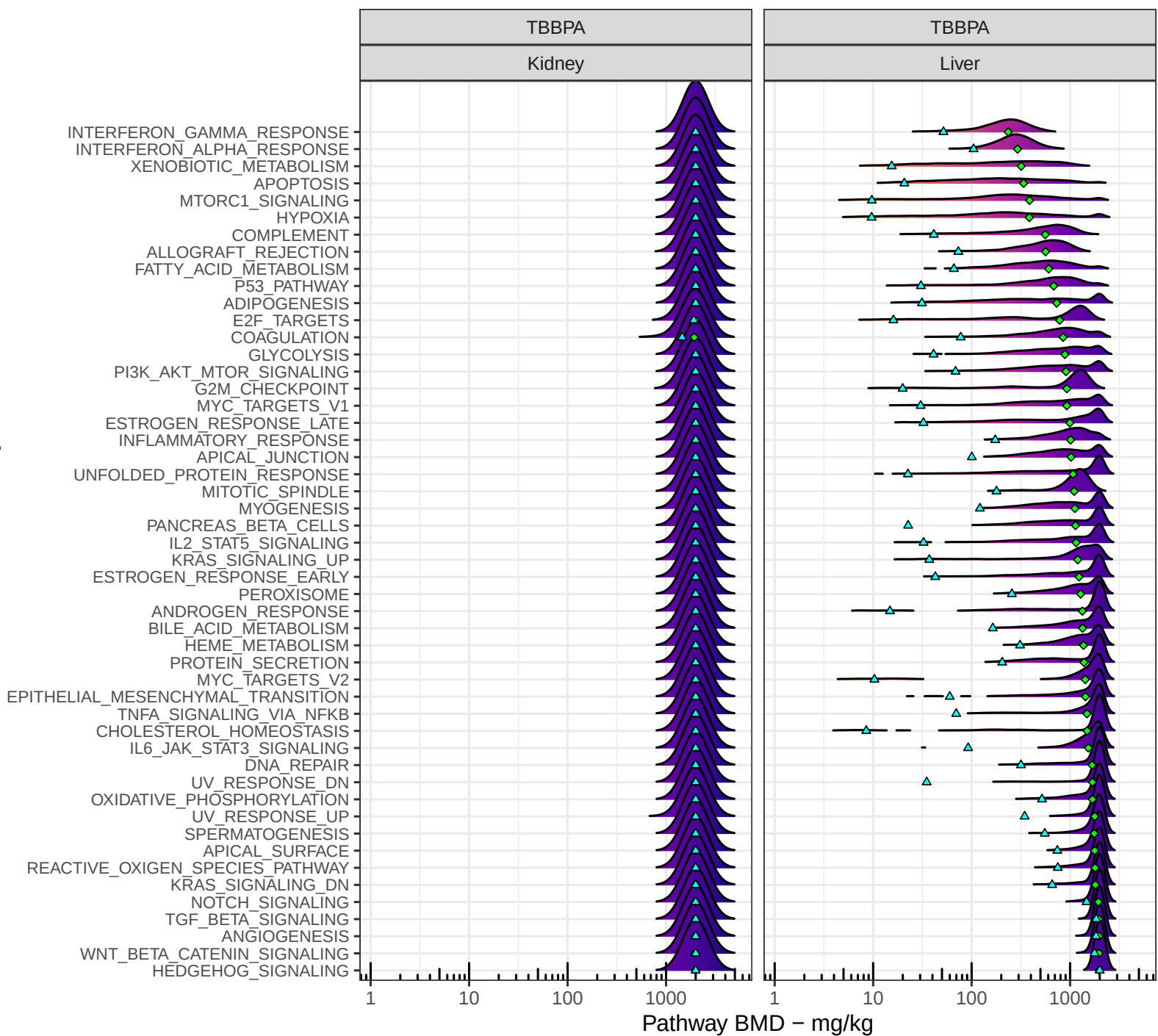


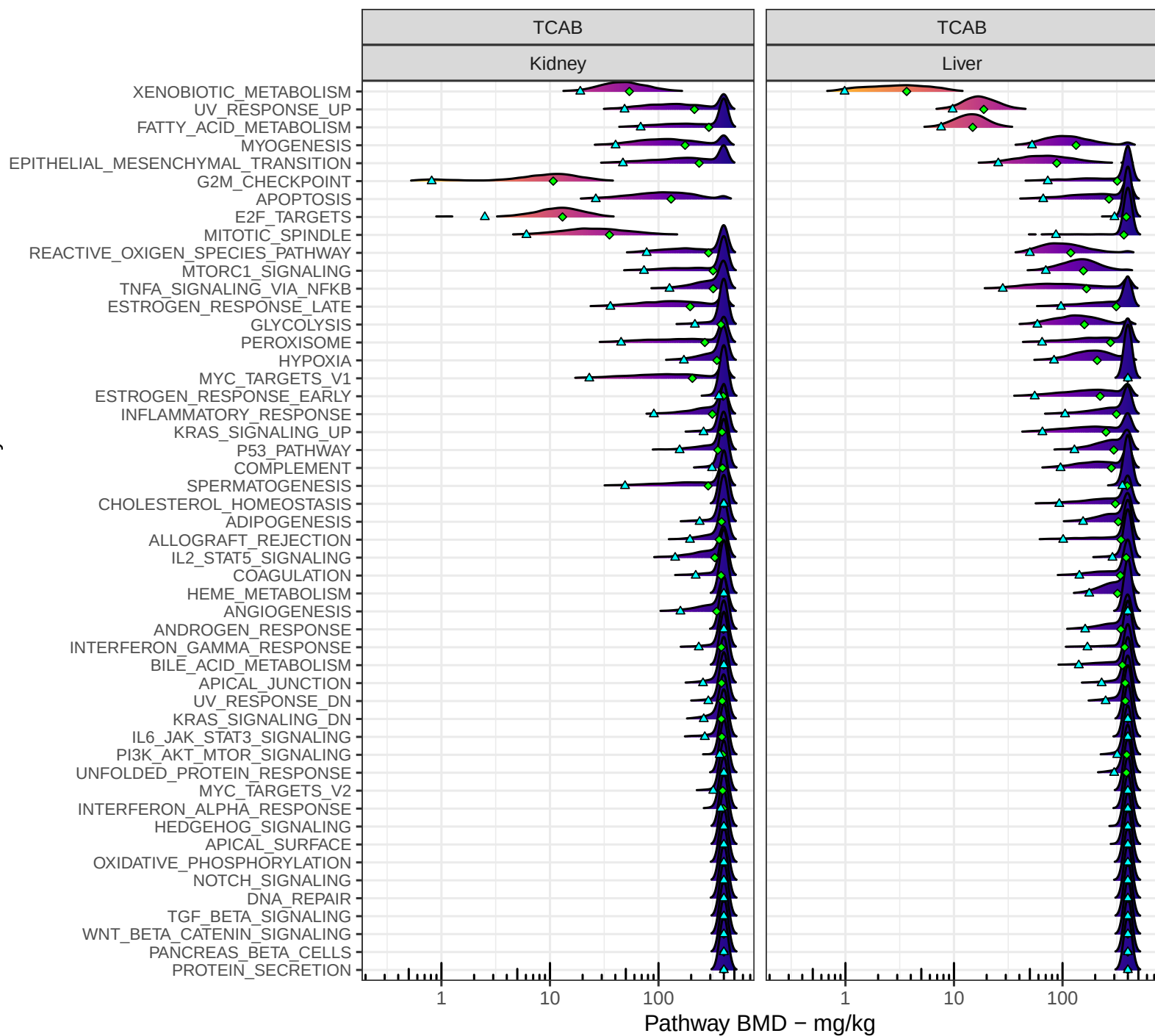


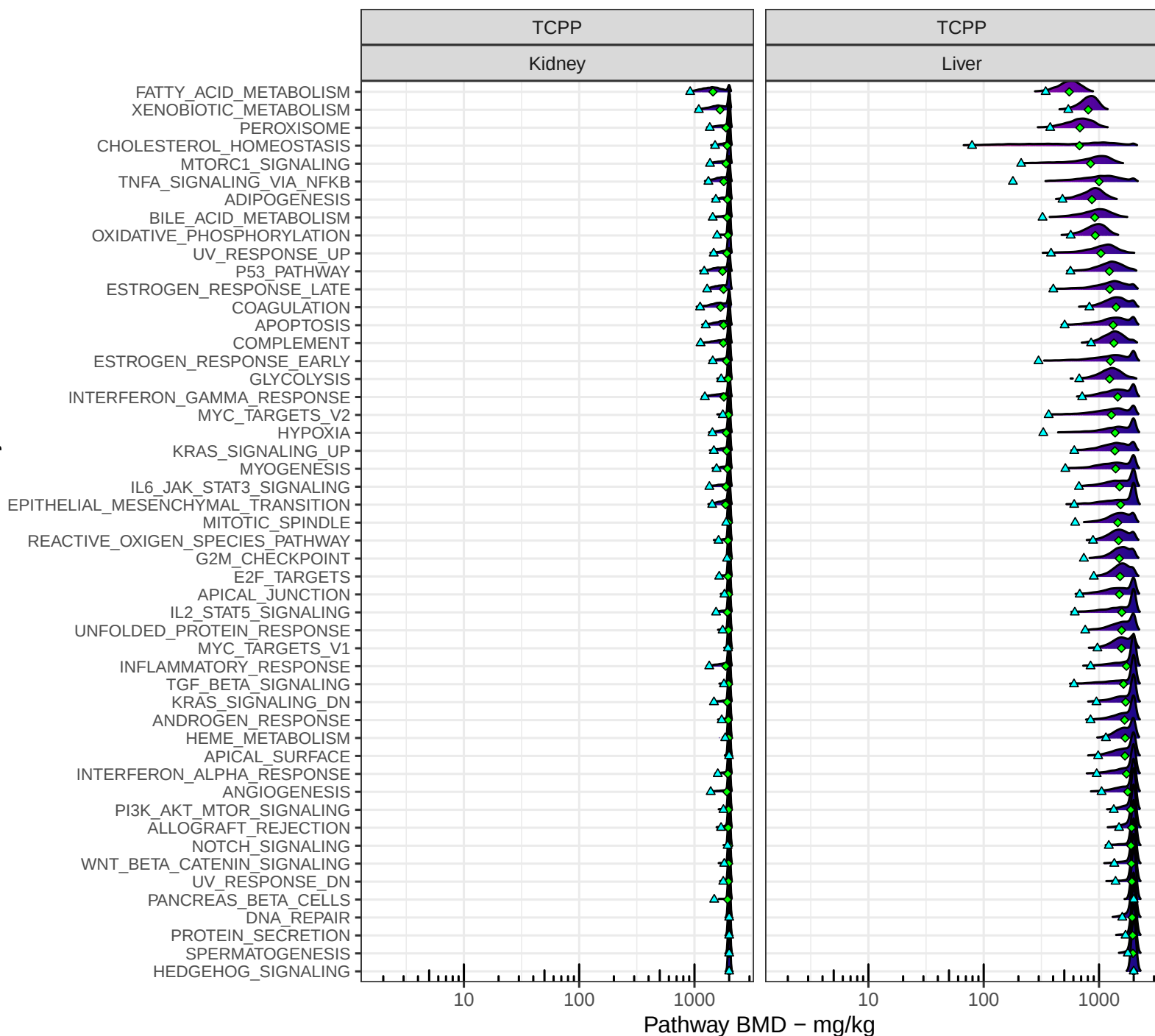










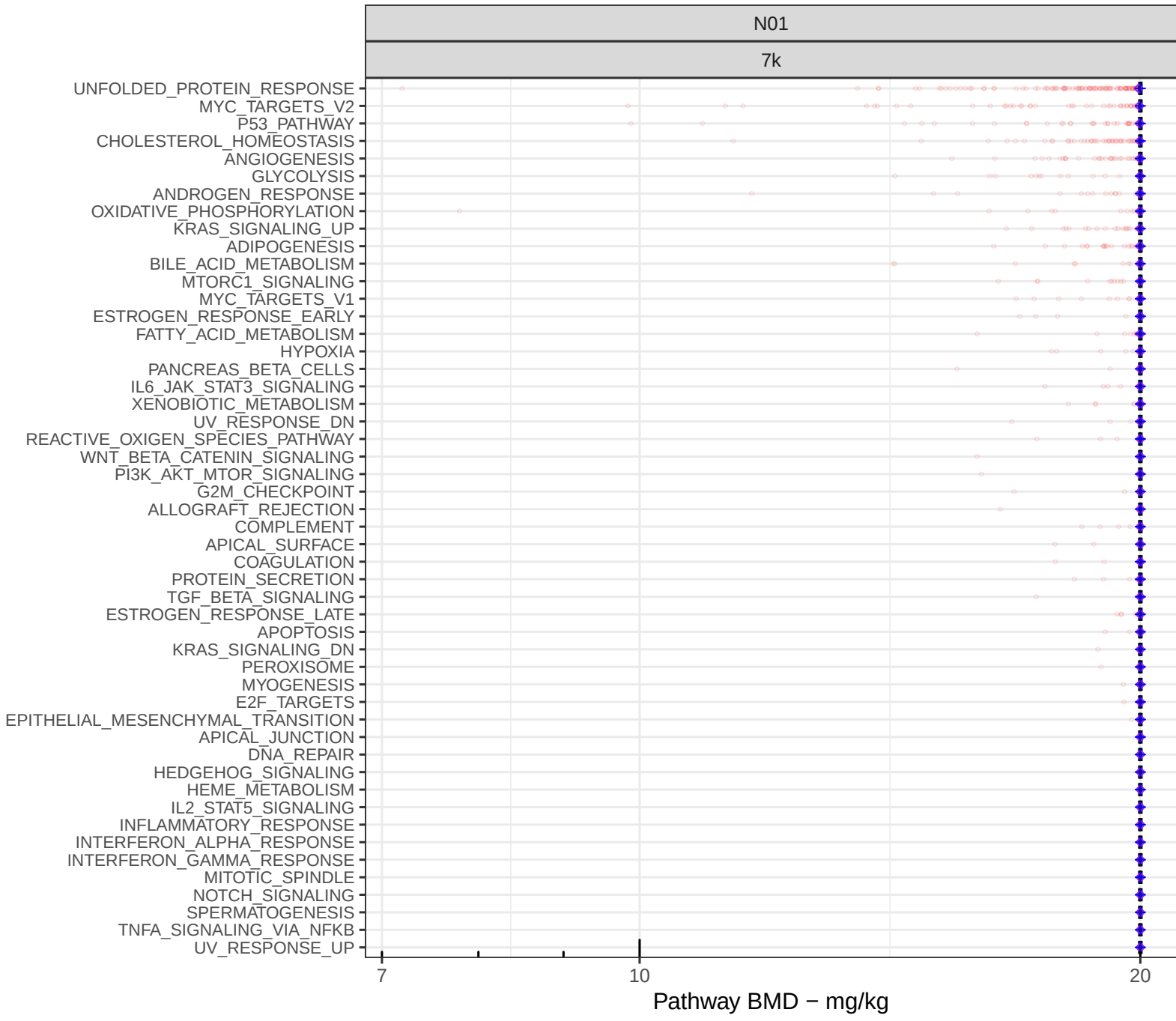


9.2 BMD posterior plots for signal-free data

Null Data generation The BGS-BMD methodology is inherently impacted by covariate gene expression, particularly when gene sets (such as the Hallmark sets) are defined in part or in totality by their coexpression. Hence, it is essential, when establishing null data sets (i.e., those made up of control samples) for the purpose of characterizing empirical false discovery, that the gene covariance structure be retained. In order to create null data sets with this characteristic, we employed male liver gene expression samples from vehicle-treated animals used in four studies that were carried out in parallel (1, 2, 3, 4). Each study had 10 control animals, hence 40 samples in total. All 40 samples were normalized using log2 (counts per million+) normalization, as described in the original manuscripts detailing the data. The normalized data were then batch corrected based on the study using an ANOVA batch correction function in Partek Genomics Suite 7.0. The batch-corrected samples were visualized using PCA to ensure the removal of study-associated batch effects. The 40 samples were then randomly selected to generate null data sets with 8 dose levels (with 1/2 dose spacing) and a zero dose control, each with 4 vehicle control samples. Five null data sets were generated in total and used for the empirical false discovery analysis in BGS-BMD. Original reports can be found at Auerbach et al. [2023a,b,c,d].

N01

Hallmark Pathway



N02

Hallmark Pathway



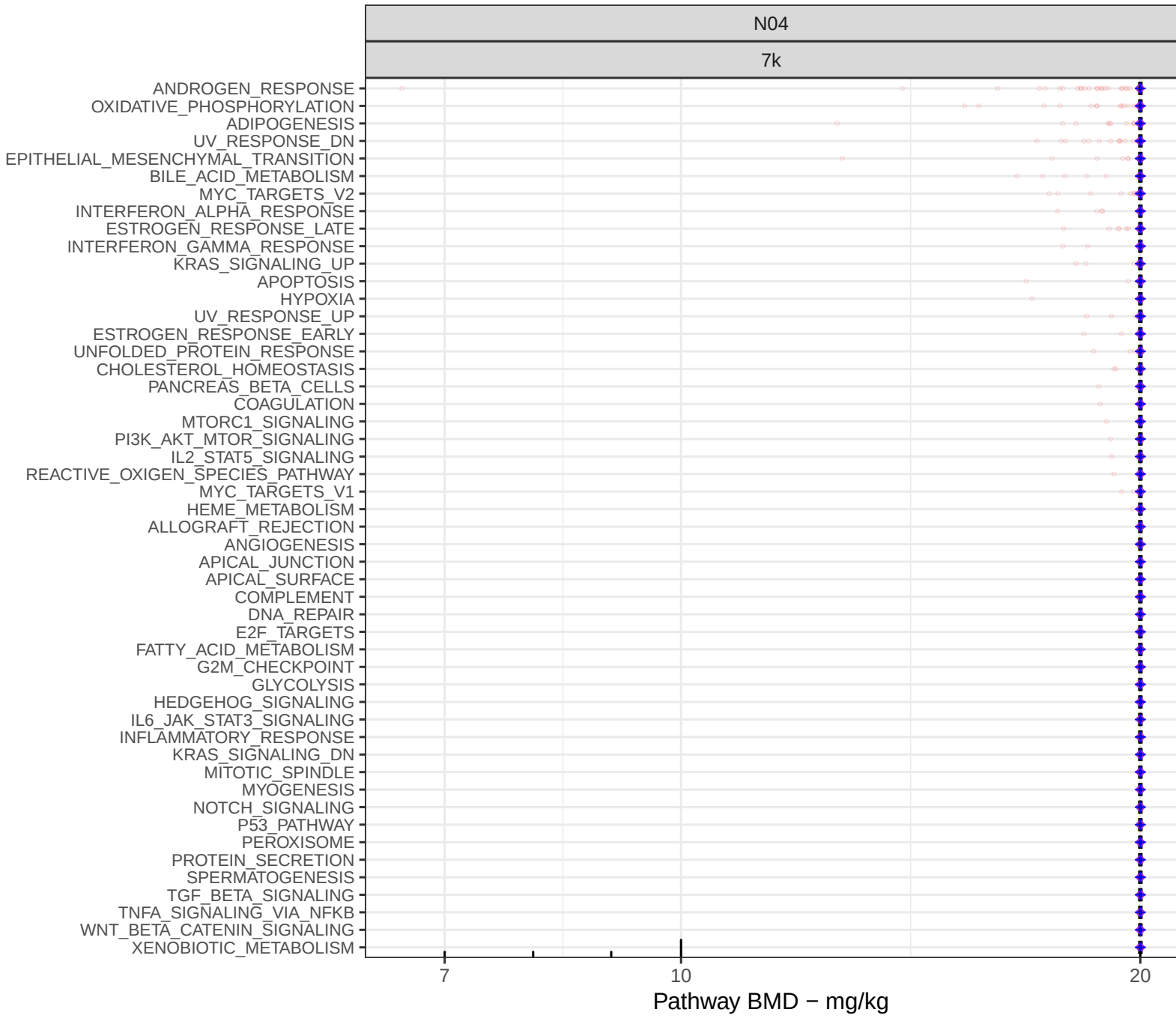
N03

Hallmark Pathway



N04

Hallmark Pathway



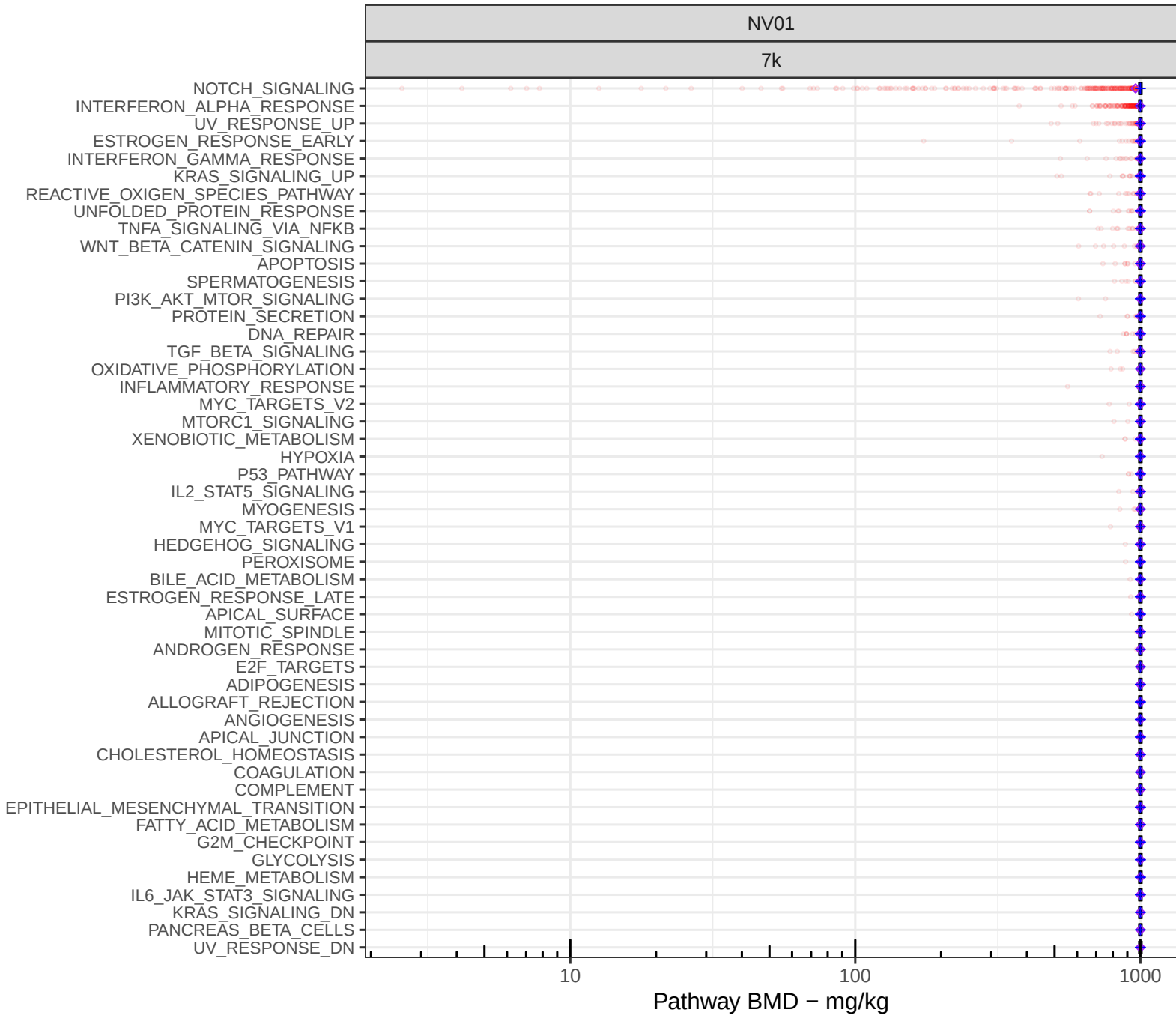
N05

Hallmark Pathway



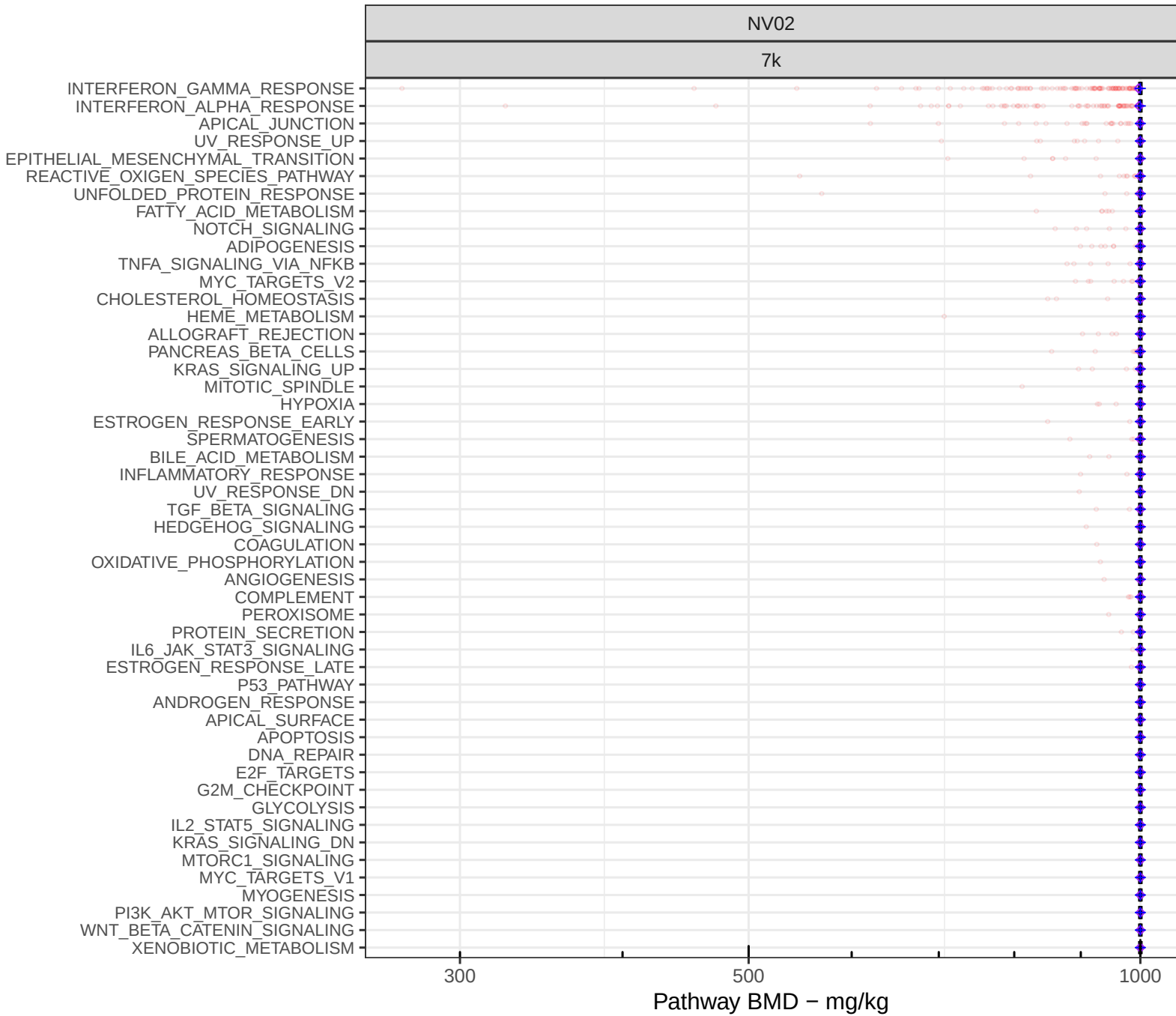
NV01

Hallmark Pathway



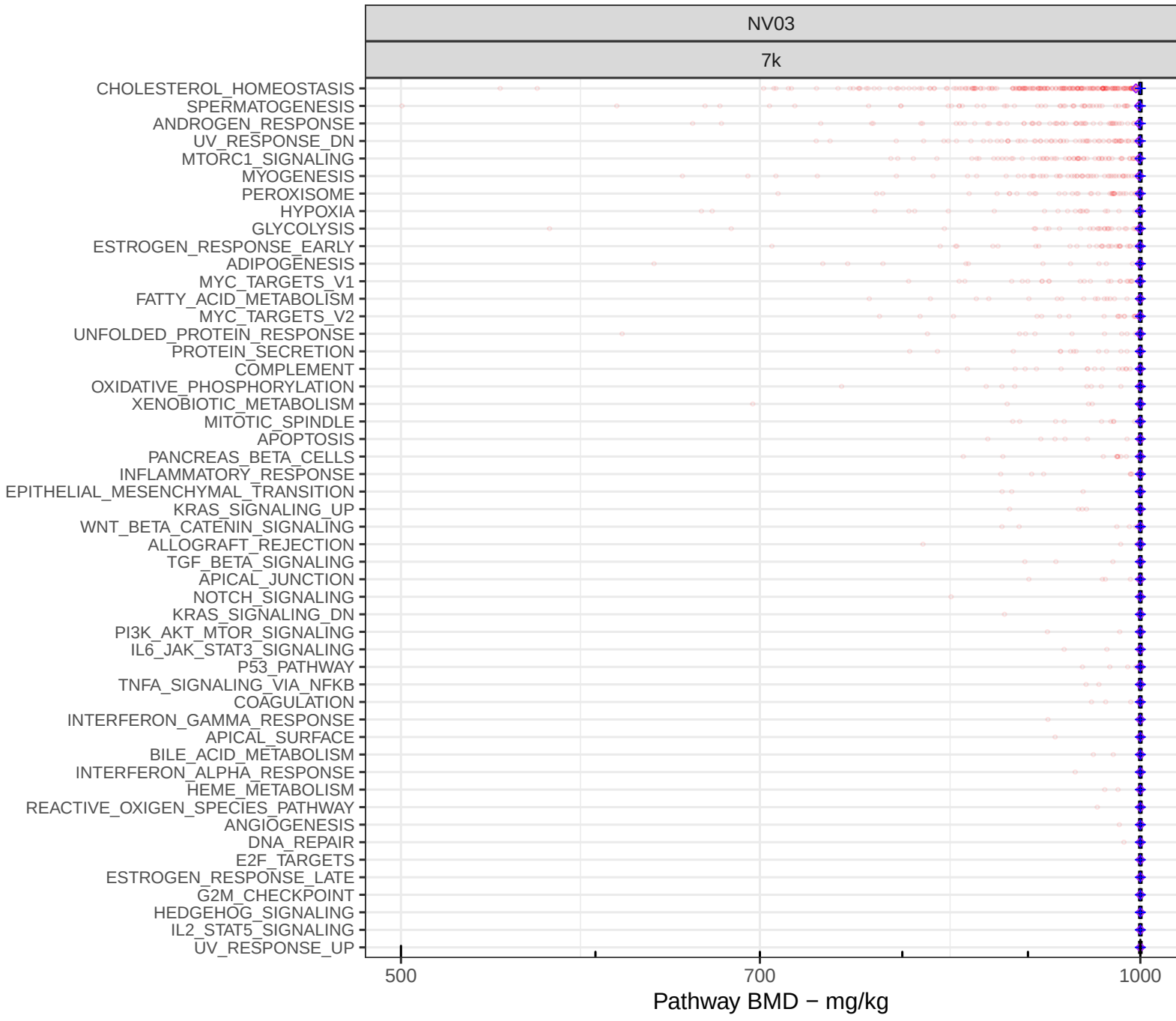
NV02

Hallmark Pathway



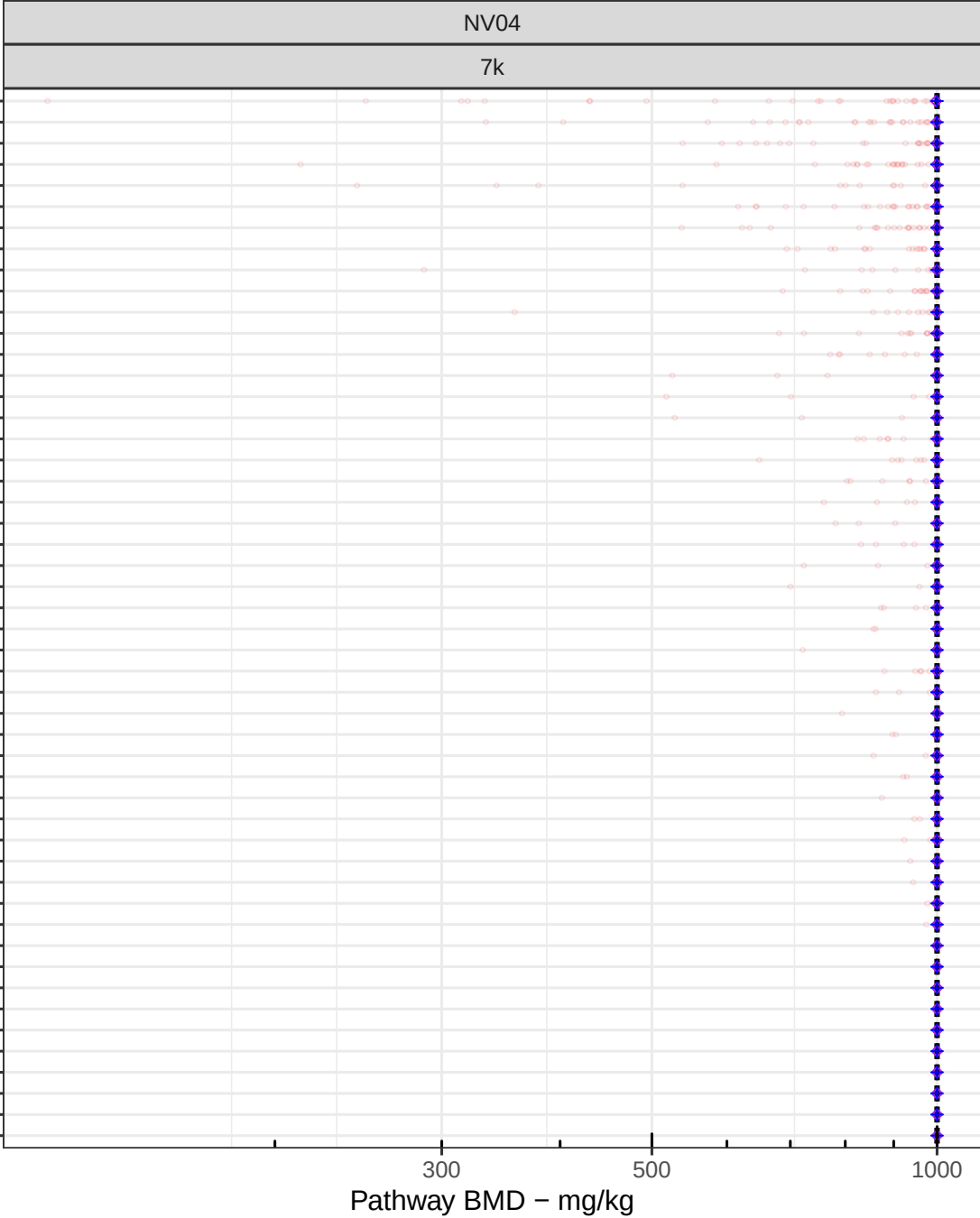
NV03

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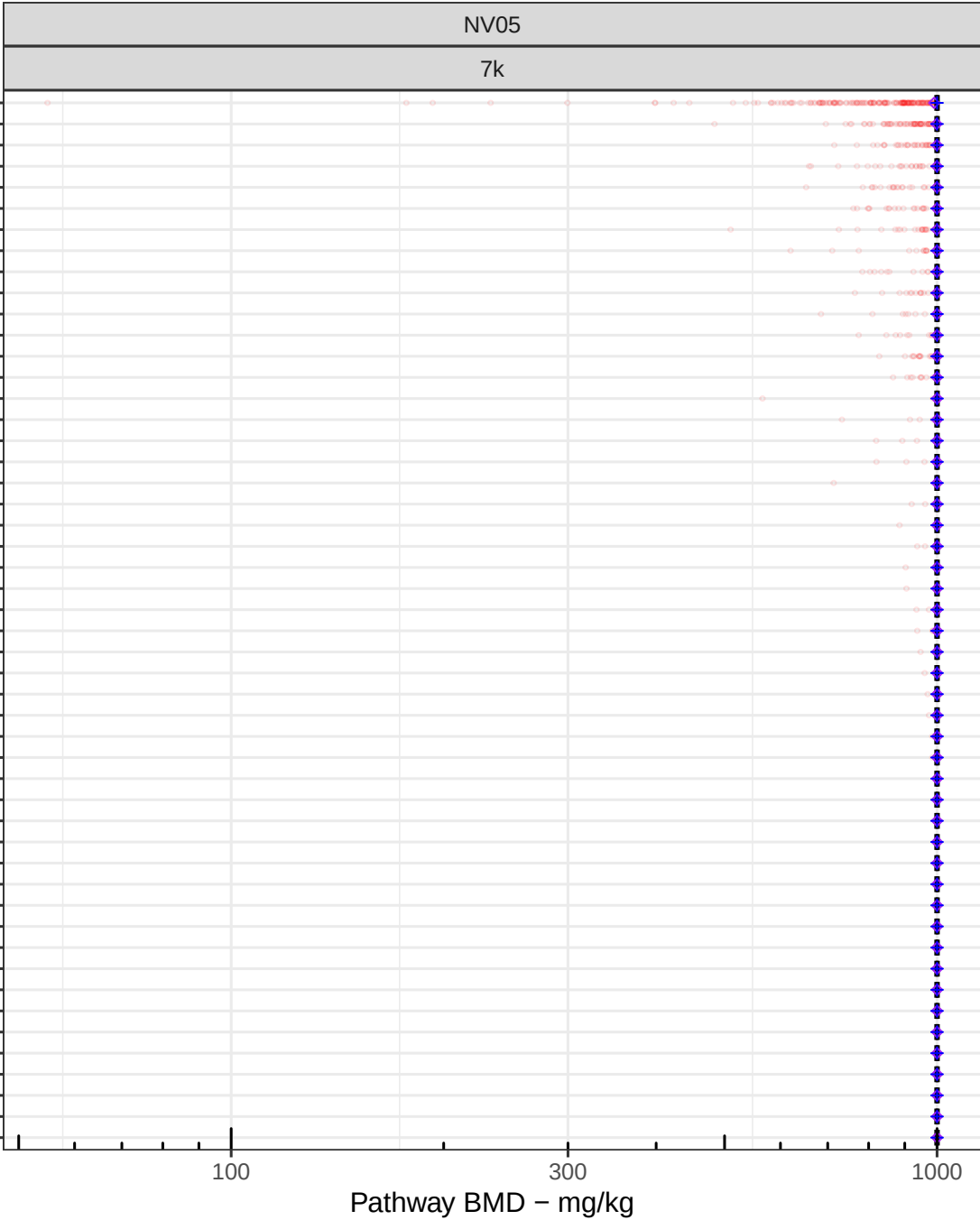
NV04

Hallmark Pathway



NV05

Hallmark Pathway



10 Apical PoD Values for Gwinn et al Study chemicals

Doses are mg/kg

Chemical	Non-neoplastic			Neoplastic			Chronic (or subchronic) Study Maximum Dose	Source/notes
	Endpoint	BMD	BMDL	Endpoint	BMD	BMDL		
ACR	Peripheral nerve (sciatic) axon degeneration	0.61	0.43	Thyroid gland follicular cell adenoma or carcinoma	1.45	0.89	2.71	1
BDCA	Bone marrow angiectasis	2.3	1.87	All organs malignant mesothelioma	4.1	3.32	43	1
COU	Liver necrosis	5.85	4.85	Kidney renal tubule adenoma	69.35	44.64	100	1
DE71	Liver hepatocyte hypertrophy	0.15	0.11	Pituitary gland (pars distalis) adenoma	6.68	4.53	50	1
EE2	Mammary gland alveolar hyperplasia	0.00069	0.00047	NA	NA	NA	0.0033	1
FEN	78 week Liver weight	0.1	0.052	NA	NA	NA	1000	10
FUR	Liver cholangiofibrosis	0.1	0.09	Mononuclear cell leukemia	0.47	0.29	2	1,2
HCB	Chronic nephrosis (severe)	0.59	0.35	Adrenal pheochromocytoma	1.23	0.60	2.35	1
MET	Liver eosinophilic focus	13.14	10.01	Liver hepatocellular adenoma or carcinoma	12.02	9.90	300	1
PFOA	Liver hepatocyte hypertrophy	0.5	0.41	Pancreas adenoma	0.73	0.57	8.8	1

PUL	Nose olfactory epithelium degeneration	14.02	9.47	NA	NA	NA	75	1
TCAB	Forestomach epithelium hyperplasia	2.86	2.13	Lung cystic keratinizing epithelioma	5.10	4.21	100	1
THU	Kidney mineralization	3.17	2.29	Preputial gland adenoma or carcinoma	22.89	16.57	50	1
Ginseng	Relative kidney weight	5056	4147	NA	NA	NA	5000	3
MTE	Platelet count	745	373	NA	NA	NA	4500	4
BPAF	F2 absolute dorsolateral prostate weight	39.5	27.8	NA	NA	NA	94	5
DEHP	Bone marrow hypercellularity	83.1	39.8	Pancreatic acinar adenoma or carcinoma	31.2	20.3	602/674	6, 11
TCPP	lung granulomatous focal inflammation	223.6	58.8	NA	NA	NA	1434	7
TBBPA	Total thyroxine	73.0	61.4	NA	NA		1000	8
TRI	Total thyroxine (BMD20)	14.5	7.23	NA	NA		300	9

Source/notes

1. BMDs and BMDLs are from Gwinn et al. 2020 Supplemental Material: Table 5 (non-neoplastic) and Table 6 (neoplastic); lowest BMD in each table for the chemical. Chronic (or subchronic) Study Maximum Dose are from Gwinn 20 Table 3.
2. Gwinn 2020 reports 2 different studies for furan, with maximum doses 2 and 8 mg/kg.

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3. NTP 90-day gavage study of ginseng, NTP TR 567. BMDs calculated using EPA BMDS online. BMD is > maximum dose.
 4. NTP 3-month feed study of milk thistle extract, NTP TR 565. Dose derived from feed dose in ppm and data on feed consumption.
 5. NTP modified one-generation study of bisphenol AF, NTP DART 08. Dose derived from feed dose in ppm and data on feed consumption. Maximum dose given is the highest dose for which there were confirmed pregnant females.
 6. Non-neoplastic from Gwinn et al. 2020 (see note 1). Neoplastic from analysis of NTP data in Table 6-4 of EPA document 'Scientific Studies Supporting Development of Transcriptomic Points of Departure for EPA Transcriptomic Assessment Products', [ps://doi.org/10.23645/epacomptox.25365550](https://doi.org/10.23645/epacomptox.25365550) March 2024. Maximum dose 602 is from EPA document, 674 from Gwinn 2020.
 7. From analysis of NTP data in Table 6-2 of EPA document 'Scientific Studies Supporting Development of Transcriptomic Points of Departure for EPA Transcriptomic Assessment Products', [ps://doi.org/10.23645/epacomptox.25365550](https://doi.org/10.23645/epacomptox.25365550) March 2024.
 8. NTP TR-587, 3-month gavage study of TBBPA, table F1.
 9. Zorrilla 2009, TOXICOLOGICAL SCIENCES 107(1), 56–64 (2009),doi:10.1093/toxsci/kfn225.
 10. Price SC, Hinton RH, Mitchell FE, Hall DE, Grasso P, Blane GF, Bridges JW. Time and dose study on the response of rats to the hypolipidaemic drug fenofibrate. Toxicology. 1986 Oct;41(2):169-91. doi: 10.1016/0300-483x(86)90198-8. PMID: 3764941.
<https://bmdsonline.epa.gov/analysis/5f1870e9-e7ff-4450-9da5-cbbb1fc8ae6e/Infplmez9a5f/#/output>
- BMD is very low relative to lowest non-control dose of 13 mg/kg.
11. <https://bmdsonline.epa.gov/analysis/bee05297-61b5-42a4-8008-eea48524e9f9/qunntut67p7p/#/output>
<https://www.ncbi.nlm.nih.gov/books/NBK579343/>

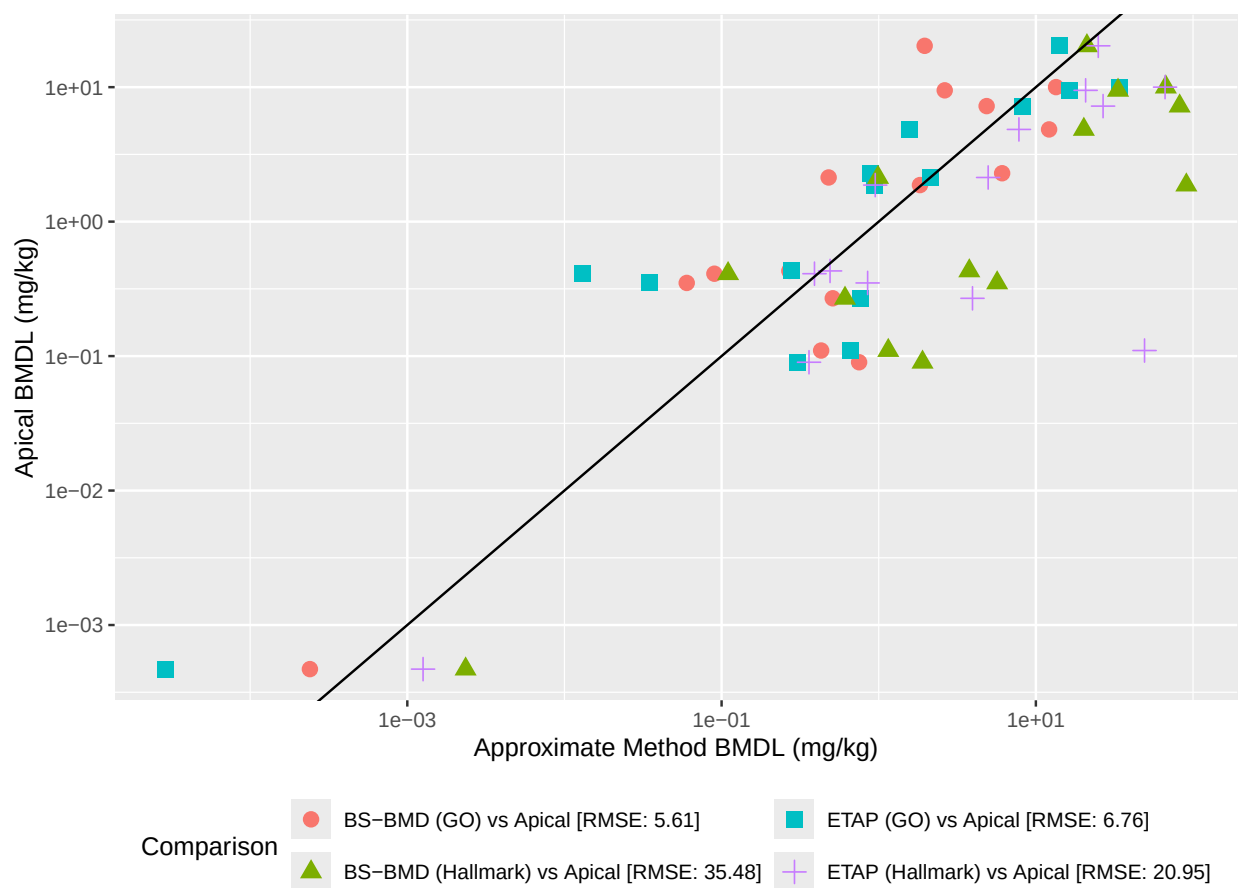


Figure 12: Four-way comparison of BMD estimates from the EPA methodology and our BS-BMD method against apical outcomes. Hallmark sets and GO terms are used for the gene sets and the lowest BMDL for each chemical across all tissues is compared to the apical outcome, if present.

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Table 7: Comparison of apical endpoint BMD to BSBMD and ETAP methods using Hallmark sets or GO terms. ApBMD refers to the apical BMD taken from Table 5 of Gwinn et al. [2020], the methods with GO refer to the lowest discovered BMD with GO sets, and the remaining columns refer to the lowest BMD found for Hallmark sets.

	Chemical	Ap. BMD	BS-BMD	ETAP BMD	BS-BMD GO	ETAP BMD GO
1	Acrylamide	0.61	10.00	0.72	8.60	0.50
2	BDCA	2.30	129.88	1.65	60.42	1.61
3	Coumarin	5.85	26.80	12.36	83.95	4.69
4	DE71	0.15	2.89	69.17	0.69	0.94
5	DEHP	31.20	30.54	48.84	196.45	22.35
6	EE	0.00069	0.0044	0.00357	0.0274	0.000241
7	Fenofibrate	2.69	0.81	6.97	10.45	2.01
8	Furan	0.10	2.40	1.05	4.29	0.55
9	Ginseng	NA	5000.00	87.90	1144.20	71.66
10	HCB	0.59	8.95	3.20	58.10	0.10
11	Methyleugenol	13.14	116.57	95.55	434.58	43.68
12	PFOA	0.50	0.13	0.72	1.47	0.05
13	Pulegone	14.02	52.19	33.98	181.55	23.92
14	TCAB	2.86	3.07	15.38	2.33	9.08
15	TCPP	14.50	554.71	71.30	890.57	33.09

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